

OmniCharacter++: Towards Comprehensive Benchmark for Realistic Role-Playing Agents

Haonan Zhang, Pengpeng Zeng, Ji Zhang, Jingkuan Song, Nicu Sebe,
Heng Tao Shen, *IEEE Fellow*, and Lianli Gao

Abstract—Existing Role-Playing Agents (RPAs), powered by large language models, are predominantly evaluated on static, text-only, dyadic conversations, which inadequately reflect the complexity of realistic human interactions involving multiple interlocutors and multi-modal communication. To bridge this gap, we propose OmniCharacter++, the first benchmark for evaluating multi-character interactions in a joint text-speech context. Specifically, OmniCharacter++ contributes: (1) a large-scale dataset comprising 10,287 characters, 118,017 multi-turn dialogues, and over one million audio responses across 8 open-world topics and 31 subfields, covering diverse multi-modal role-playing scenarios; (2) a comprehensive evaluation suite for dialogue understanding, generation quality, and perceptual naturalness; and (3) UniCharacter-7B, a unified text-speech model trained on this dataset to manage complex multi-character dynamics, ensuring both role-specific vocal fidelity and cross-participant semantic alignment. Experimental results demonstrate that UniCharacter-7B achieves more realistic and consistent role-playing responses in terms of both attractiveness and consistency, while also highlighting that OmniCharacter++ poses substantial challenges for state-of-the-art models, charting a clear path for future research. The Code is publicly available at: <https://github.com/zchoi/OmniCharacter-plus>.

Index Terms—Role-Playing Agents, Multi-Agent Learning, Text-Speech Understanding and Generation, Multi-Turn Dialogue

1 INTRODUCTION

ROLE Playing Agents (RPAs) aim to sustain coherent, persona-consistent interactions across diverse conversational contexts, *i.e.*, a social and adaptive capability that humans naturally excel at but current AI systems still struggle to achieve. In recent years, advances in AI have fueled increasing interest in enhancing these abilities, driving progress in areas such as digital companionship [1], [2], educational support [3], [4], and interactive entertainment [5].

The development of RPAs has been significantly accelerated by the emergence of Large Language Models (LLMs) [6], [7], which provide a solid foundation for simulating different personas with diverse behaviors based on predefined profiles. Generally, existing research can be categorized into three main directions. Early efforts [8], [9], [10] primarily aimed to enable agents to emulate specific personas by grounding them with explicit prompts or alignment training to adhere to the character knowledge or behavior patterns. While effective in traditional dialogue generation, these methods often fall short in capturing complex, human-like interactions. To address these limitations, subsequent studies focused on human simulation for RPAs by incorporating the concept of ‘social intelligence’ that simulates beliefs, intentions, and perspectives of humans via LLMs, making it more practical and valuable. Conversely, it also contributes to advancing textual modes like LLMs and language agents by providing a testbed to examine whether

theory-of-mind abilities genuinely exist [11], [12]. Beyond that, RPAs also facilitate the creation of diverse, large-scale synthetic data for LLMs training [13]. Leveraging varied persona patterns, RPAs can produce linguistically and situationally textual data, enhancing LLMs’ generalization and adaptability.

Despite significant advances, existing approaches still face inherent limitations, where they heavily rely on static interactions with simplified dialogue structures. This ideal setup, however, may rarely reflect the realities of real AI interactive systems. Indeed, humans’ ability to achieve and balance complex, multifaceted interactions is a critical symbol of social intelligence. A successful communication among human beings requires not only understanding the intentions of the individual they are talking to, but also considering the potential advantages or conflicts especially when multiple parties are involved. For example, even a simple social situation like deciding where to eat with friends requires balancing different needs, *i.e.*, one person’s craving for spicy food, another’s diet restrictions, and a third’s tight budget. Moreover, effective communication often needs multi-modal perceptions for real-time tracking of dialogue scenarios and the subsequent statements. In detail, people exhibit diverse speaking styles and emotional expressions, and the same sentence—or even the same sentence in varying contexts—may carry very different meanings. Even subtle variations in dialogue, such as a slight change in tone, can signal markedly distinct intention. Compared to human interactions, while the most advanced RPAs exhibited impressive performance in certain settings, their ability has yet to be ascertained under real-world contexts.

To bridge this gap, we introduce a new benchmark called **OmniCharacter++**, the first benchmark designed to investigate large-scale, multi-character interactions in a multi-

- Haonan Zhang, Pengpeng Zeng, Jingkuan Song, and Heng Tao Shen are with Tongji University, China.
- Ji Zhang is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, China.
- Nicu Sebe is with the Department of Information Engineering and Computer Science, University of Trento, Italy.
- Lianli Gao is with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, China.

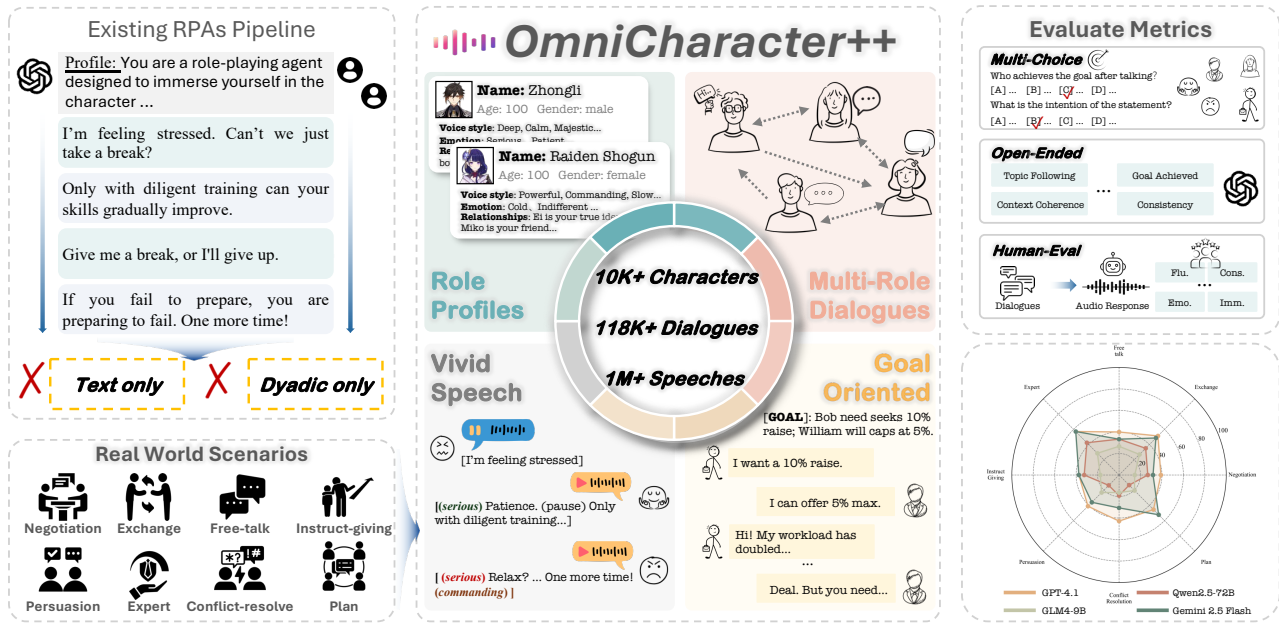


Fig. 1: Overview of OmniCharacter++ Benchmark. Compared to previous RPA benchmarks that primarily focus on text- and dyadic-only dialogues, OmniCharacter++ comprises realistic multi-character interactions across diverse open-world topics with text-speech co-driven exchanges, featuring 10K characters, 118K dialogues, and 1M audio responses, along with a comprehensive evaluation suite for benchmarking current models.

modal environment shown in Fig. 1. Unlike previous benchmarks that focus on static, text-only manner with dyadic dialogue structures, OmniCharacter++ captures the complexity and diversity of human-like interactions. Specifically, it supports various dialogue scenarios, ranging from two-person to group conversations, with over 118,017 multi-turn dialogues across 10,287 unique characters, along with more than one million synthesized audio responses, covering 8 major open-world topics with 31 subfields. Moreover, a comprehensive evaluation suite is introduced to assess the capabilities of the above multi-modal multi-character role-playing from various perspectives. Particularly, it includes: (1) **Context Understanding**: given a dialogue history and a character's intent, models are required to answer content-related questions to demonstrate an accurate grasp of the conversation; (2) **Generation Ability**: refers to measuring the model's capacity to produce coherent and contextually appropriate responses across four aspects, *i.e.*, topic following, task success, consistency, and context coherence; and (3) **Human Perception**, where human experts evaluate the synthesized audio responses with respect to six metrics: fluency, consistency, emotional expression, clarity, appropriateness, and immersion, ensuring naturalness and fidelity in speech-based interactions. Further, to tackle the challenges of complex multi-character conversation, we introduce a novel baseline model called *UniCharacter*. It serves as a unified speech-language dialogue agent that generates both text and speech responses aligned with the intent and persona of various characters. In addition, UniCharacter employs an Emotion Preference Learning strategy (EPL), allowing discriminative control for more expressive reactions. We further enhance its dialogue abilities by involving a Role-contextual Dialogue Adaptation (RDA) to facilitate flexi-

ble interaction with new characters and scenarios without additional fine-tuning. We evaluate state-of-the-art models on our OmniCharacter++ benchmark. Our findings reveal that even top-performing models struggle to follow predefined goal instructions in topic-driven conversations, and this limitation is particularly pronounced in multi-character settings. Effectively perceiving and maintaining nuanced contextual and role-specific behaviors across diverse characters remains a critical challenge, highlighting the need for further research. To sum up, we make the following contributions in this work:

- To the best of our knowledge, OmniCharacter++ is the first benchmark for realistic multi-character role-playing across text-speech modalities, offering diverse data corpus and a comprehensive evaluation framework that serves as a testing ground for investigating role-playing behaviors and capabilities through systematic, multi-perspective assessment.
- We propose UniCharacter, a novel speech-language model for realistic role-playing. It learns nuanced vocal styles through emotion preference learning from contrastive audio pairs and adapts to OOD character conversations via a retrieval mechanism that requires no retraining. UniCharacter offers a versatile and scalable framework, establishing a strong baseline for future research on RPAs.
- We have evaluated various methods and UniCharacter on OmniCharacter++ under three different tasks. Extensive experiments reveal that a wide range of models encounter significant performance gaps, *e.g.*, with none reaching >55% average score on the context understanding task, highlighting the current models' shortcomings in realistic role-playing, and

TABLE 1: **The comparison with existing RPAs benchmarks considers multiple dimensions:** the total number of characters (**#Characters**); the number of samples (**#Samples**), where each sample denotes either a dialogue or a question, and * indicates the count of questions; the supported languages (**Language**); the source domains of the data (**Data Domain**); the involved modalities (**Modality**); the annotation methodology (**Anno.**, with A/M denoting automatic/manual annotation); whether the dialogue covers multi-character conversations (**Multi-Character Conv.**), whether the conversation is conducted around a certain topic (**Topic**), and whether the conversation aims at achieving a certain goal (**Goal-oriented**).

Benchmark	#Characters	#Samples	Language	Data Domain	Modality	Anno.	Multi-Character Conv.	Topic	Goal-oriented
Character-LLM [10]	9	14,300	EN	Encyclopedia	Text	A	✗	✓	✗
ChatHaruhi [15]	32	54,726*	EN	Books, Games, Movies	Text	A	✗	✗	✗
HPD [16]	-	1,191	EN, ZH	Books	Text	A+M	✗	✗	✓
CharacterGLM [17]	250	1,034	ZH	Books, Scripts	Text	A+M	✗	✓	✗
PIPPA [18]	1,254	25,940	EN	Character.ai-Users	Text	A+M	✗	✗	✗
RoleEval [19]	300	6,000*	EN, ZH	Encyclopedia	Text	A	✗	✓	✓
RoleLLM [9]	100	140,726*	EN, ZH	Scripts	Text	A+M	✗	✗	✗
CharacterEval [14]	77	11,376*	ZH	Books, Scripts	Text	A+M	✗	✓	✓
DITTO [20]	4,002	36,662*	EN, ZH	Encyclopedia	Text	A	✗	✓	✗
RolePersonality [21]	46	32,767	EN	Personality Tests	Text	A+M	✗	✗	✗
MORTISE [22]	190	17,835	EN, ZH	Encyclopedia, Other Dataset	Text	A	✗	✓	✗
Cross-MR [23]	126	445*	EN	Literature-Analysis	Text	A+M	✗	✓	✗
SocialBench [24]	500	30,800*	EN, ZH	Books, Movies	Text	A	✗	✓	✓
TimeChara [25]	14	10,895	EN	Books	Text	A	✗	✗	✗
LifeChoice [26]	1,401	1,401*	EN	Literature-Analysis	Text	A+M	✗	✓	✗
InCharacter [27]	32	18,304*	EN, ZH	Personality-Tests	Text	A+M	✗	✗	✗
MMRole [28]	200	30,800*	EN, ZH	Encyclopedia	Text+Image	A+M	✗	✗	✗
OmniCharacter [29]	20	10,172	EN, ZH	Games	Text+Audio	A+M	✗	✗	✗
OmniCharacter++	10,287	118,017	EN, ZH	Open-Domain	Text+Audio	A+M	✓	✓	✓

the need for further advancements. UniCharacter achieves superior performance compared to its competitors in both the OmniCharacter++ test set and the OOD role-playing benchmark CharacterEval [14], demonstrating its effectiveness and generalizability.

Differences from the conference version. This work stems from our ACL 2025 paper, *i.e.*, OmniCharacter [29]. The significance of this work lies in the following key contributions:

- **Dataset Expansion:** OmniCharacter focuses on role-playing within the game domain, primarily featuring two-party dialogue. In contrast, OmniCharacter++ extends this to open-domain role-playing, supporting both two-party and multi-character (up to four characters) dialogue based on topics and goal instruction. In terms of scale, OmniCharacter++ represents a remarkable expansion from characters (20 *vs.* 10,287), dialogues (10,072 *vs.* 118,017), and audio responses (370+ hours *vs.* 3900+ hours), establishing a comprehensive benchmark for studying realistic role-playing interactions.
- **Comprehensive Evaluation Framework:** To better testify role-playing abilities in multi-character and multi-modal scenarios, OmniCharacter++ extends OmniCharacter by introducing a new evaluation framework. It comprises three key components: (1) Context Understanding, assessing role-related comprehension through challenging QA tasks; (2) Generation Ability, measured by four sub-metrics capturing response quality; and (3) Human Perception, a fine-grained assessment across six dimensions reflecting the fidelity of character expressions.
- **Model Advancements:** We present UniCharacter-7B, a model that incorporates two key innovations over its predecessor, OmniCharacter-7B. (1) It integrates emotion-preference learning (EPL) for controllable emotional expressivity and role-contextual dialogue adaptation (RDA) for theme-aligned, in-context interactions without additional training. (2) Its training on the expansive OmniCharacter++ dataset enables

it to adeptly handle not only dyadic interactions but also complex multi-party conversations.

- **Insightful Findings:** Beyond reporting improvements—where UniCharacter surpasses the previous model on most metrics—the evaluation suite itself reveals critical limitations of current models on OmniCharacter++, showing that even the strongest models still struggle with nuanced role consistency and expressive robustness. Our dataset along with these findings underscores the need for further improvement in handling realistic character behaviors, shedding light on future role-playing agent development.

2 RELATED WORK

We briefly review recent advances that are most related to our work in the context of RPAs and text-speech models.

2.1 Role-Playing Agents

Recent advancements in large language models (LLMs) have enabled the development of Role-Playing Agents (RPAs), which emulate personas through multi-turn dialogue. Most existing approaches [17], [30] focus on individual character simulation by conditioning LLMs with character-specific data, either via fine-tuning or prompt-based adaptation. For instance, [16], [31] construct agents by extracting dialogues from franchise scripts. Moreover, subsequent studies [17], [32] extend this paradigm to large-scale persona collections spanning hundreds of diverse characters. [30] reduces hallucination by adjusting parametric knowledge influence based on a pre-calibrated confidence threshold. [10], [33] seek to train an agent with profile and experience perception, replacing limited prompts for LLM instruction. Despite these developments, our prior work [29] reveals that most existing RPAs remain limited to text-only interactions, lacking support of roles' essential voice traits, such as tone, emotions, and pauses, which are crucial for realistic role-play.

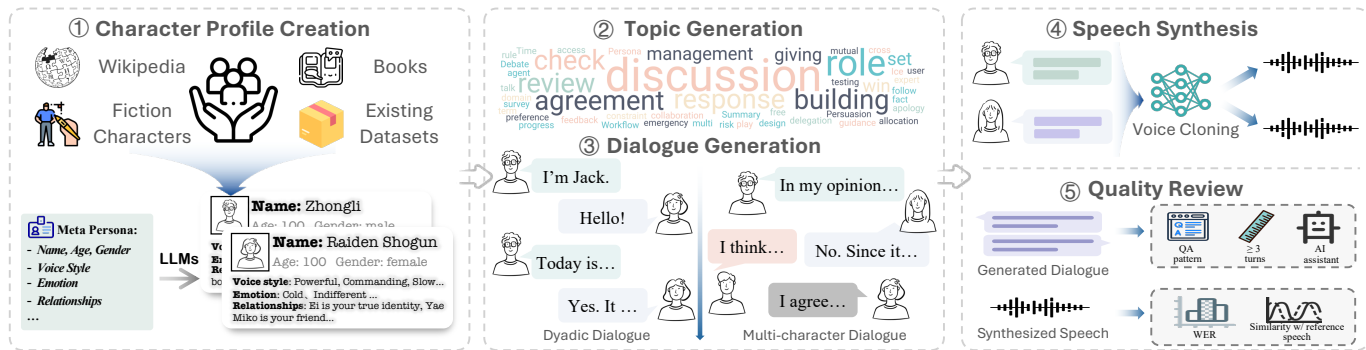


Fig. 2: **The construction of OmniCharacter++.** The pipeline collects character profiles from multiple sources and refines them through LLMs to ensure both quality and contextual relevance. Next, hierarchical topics are defined, and dialogues are generated based on the provided contexts such as profiles, scenarios, goals, and topics. Further, character-specific audio is annotated for each sentence using voice cloning. Finally, a rigorous quality review process is conducted to ensure the accuracy and naturalness of both the dialogues and speech annotations.

Beyond leveraging powerful LLMs for RPAs, recent work has increasingly focused on constructing character-centric datasets [8], [31], [34] and systematically evaluating agent capability and behavioral consistency [27]. Several studies [14], [24], [35] propose structured benchmarks that move beyond surface-level dialogue generation. For instance, [14] introduces a Chinese benchmark with high-quality multi-turn dialogues and multi-dimensional evaluation, while [24] presents the first framework for assessing social intelligence in RPAs at both individual and group levels. Although existing RPA benchmarks offer multi-dimensional evaluation protocols, their reliability remains limited. Most approaches rely on open-ended generation assessed by LLM-as-a-judge or reward models, leading to two fundamental issues: (1) stochastic generation yields inconsistent outputs under identical inputs, undermining evaluation stability; and (2) results are highly dependent on the judge model, whose inductive biases may deviate from human preferences. To address these limitations, OmniCharacter++ proposes a structured hybrid evaluation framework. Context understanding is formulated as a multiple-choice task with a Circular Evaluation Strategy [36] to reduce stochasticity, while conversational ability is assessed via a goal-driven protocol covering instruction adherence, goal completion, persona consistency, and coherence. Human expert evaluation is further incorporated to mitigate judge bias. Overall, the framework provides more reliable and interpretable evaluation for RPAs.

2.2 Speech-Language Models

Progress in speech-language modeling has substantially enhanced the fluency, coherence, and naturalness of human-machine interactions [37], [38], [39], [40], [41]. With the advancement of language models in the natural language processing field, early work attempted to enable language models to generate both speech tokens and text tokens via a decoder-only architecture [42], [43], [44]. With the development of large language models (LLMs), more recent studies have integrated speech capabilities into LLMs by either adding speech tokens to the text vocabulary or incorporating a speech encoder before the LLMs such as

TABLE 2: The statistics analysis of OmniCharacter++.

Set	Dialogue	#Characters	Avg. Turn/Conv.	#Samples	#Speech Hours
Train	Dyadic	10,277	10.00	88,474	2867.94
	Multi-Character		15.05	29,543	1051.66
Test	Dyadic	10	9.89	185	6.96
	Multi-Character		16.72	334	15.20

SpeechGPT [41] and AudioPaLM [45]. Beyond that, some methods fine-tune LLMs for speech understanding such as LLaMA-Omni [37] and GLM-4-Voice [40], enabling them to perform general speech tasks with low response latency. Additionally, models like Moshi [46] and OmniFlatten [47] handle real-time interactions by managing simultaneous speech and text streams. Despite progress, most existing speech-language models explore general audio-language capabilities. Compared with the above methods, we focus on modeling the unique traits of different roles in RPA tasks for a more realistic human-machine interaction.

2.3 Multi-Agent Simulation

Multi-agent simulation offers a powerful paradigm for modeling complex human behaviors in interactive environments [6], [48], [49], [50], [51]. Unlike isolated agents, multi-agent systems enable dynamic interaction, communication, and coordination, giving rise to emergent behaviors that better reflect real-world social dynamics. These agents capture diverse aspects of human cognition and emotion, including goals, motivations, and social relationships, which shape their decision-making processes. Recent works leverage this paradigm for evaluation and robustness. For example, ChatEval [49] employs multi-agent debate for collaborative assessment, while [50] introduces a general framework to enhance reliability in complex tasks. Stable Alignment [52] further explores scalable training via simulated social interactions to improve alignment and robustness. However, existing approaches remain largely limited to text-only settings and struggle to model multi-agent, multi-modal interactions with rich emotional and behavioral dynamics. Compared to dyadic dialogue, multi-agent scenarios are more realistic yet significantly more challenging, due to scarce data and substantial gaps from real-world interactions.

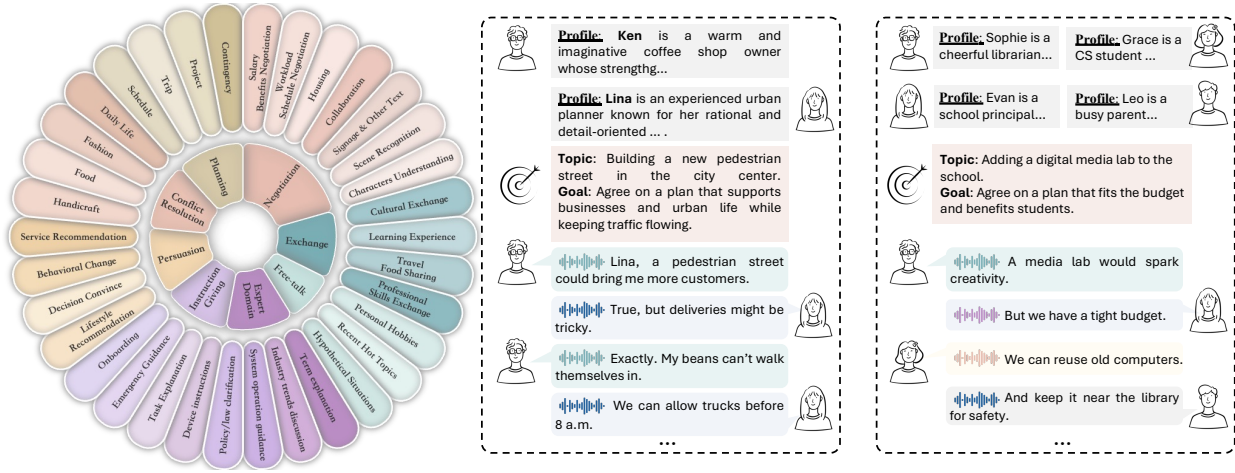


Fig. 3: **Illustration of OmniCharacter++**. Left: topic categories. Our benchmark covers 8 key topics and 31 sub-class types. Right: dialogue samples in our dataset. It contains both dyadic and multi-character conversations, accompanied by rich annotations (topic, goal, and speech) that enable agents to engage in interactions with consistency and a human-like manner. Note that the above dialogues have been slightly simplified for clarity and conciseness.

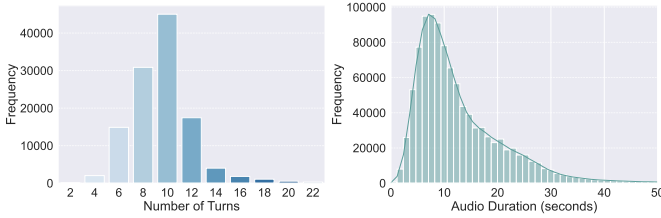


Fig. 4: **Left**: Distribution of dialogue turns. **Right**: Distribution of audio durations.

3 OMNICHARACTER++ BENCHMARK

To facilitate the study of RPAs in a more realistic and interactive environment, we propose OmniCharacter++, a large-scale benchmark for multi-modal and multi-character role-playing. In this section, we describe the data construction of OmniCharacter++ in Section 3.1 and introduce the dataset statistics in Section 3.2. The comparison with the existing RPAs benchmarks is depicted in Tab. 1.

3.1 Dataset Construction

The construction of the OmniCharacter++ involves five main steps: character profile creation, topic generation, dialogue generation, speech synthesis, and quality review. The dataset construction pipeline is demonstrated in Fig. 2.

① **Character Profile Creation**. To ensure comprehensive character diversity, we organize the character space into three primary persona categories following prior work [53]: (1) *Demographic Persona* (e.g., mathematician, baseball enthusiast, millennials), describing groups sharing common attributes and leveraging statistical regularities in LLMs; (2) *Character Persona* (e.g., Napoleon, Harry Potter, Elon Musk), corresponding to well-known real or fictional figures that require structured world knowledge; (3) *Individualized Persona*, representing personalized or synthetic profiles with unique and evolving characteristics. We source characters from diverse domains, including Wikipedia, novels,

TABLE 3: Artifact analysis of OmniCharacter++ dataset.

Dataset	Distinct-1	Distinct-2	Self-BLEU ↓
CharacterEval [14]	0.061	0.243	0.36
OmniCharacter++	0.087	0.342	0.214

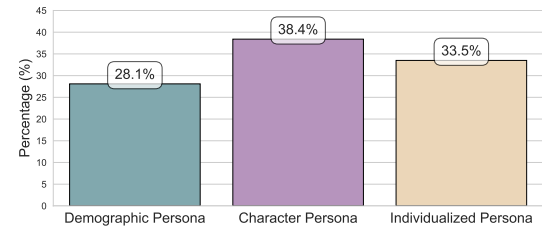


Fig. 5: **Person category distribution in OmniCharacter++**.

scripts, online platforms (e.g., CharacterAI¹, Fandom²), and LLMs (GPT-4o, Doubao) prompting. Each character is associated with a structured profile consisting of five components: a brief introduction, personality traits, speaking style, emotions, and interpersonal relationships. To ensure consistency and theoretical grounding, personality and emotion attributes are defined based on established frameworks (e.g., primary personality traits (Positive, Neutral, and Negative Traits) [54] and Plutchik's wheel of emotions [55]). After constructing the initial character priors, we further employ GPT-4o to summarize meta-information and expand each profile, improving both diversity and completeness. To ensure reliability, all generated profiles undergo manual sampling-based quality checks, mitigating hallucination and maintaining role consistency.

② **Topic Generation**. Human dialogues are often driven by implicit or explicit social and functional objectives, such as seeking assistance, negotiating outcomes, exchanging information, or interpreting intentions. To capture this diversity,

1. <https://beta.character.ai/>
2. <https://www.fandom.com/>

⑤ **Quality Review.** To ensure the quality of the constructed dataset, we manually filter samples based on several criteria. For the training set, we conduct sampling-based human evaluation and iteratively refine the prompts based on feedback to improve data quality. The test set is fully human-validated to ensure reliable evaluation. In detail, for text data, we perform data filtering by retaining samples that follow the AB-AB (user-role or role1-role2) pattern for two-party dialogue and ABCD-ABCD (role1-4) pattern for multi-character dialogue. Additionally, we preserve dialogues that are longer than three turns, filtering out those that are shorter. Furthermore, we remove data that mimics the style of general AI assistance such as ‘*I am a helpful AI assistant...*’, as well as unnecessary explanatory prefaces and postfaces. For speech data, we first use the Whisper-large-v3 [57] to convert the synthesized audio into text and compute the WER metrics with the corresponding text, where the WER greater than 10 is rejected. Then, we use WavLLM [58] to compute the similarity between the synthesized audio and reference audio, where the similarity less than 0.8 threshold is also rejected.

Tab. 2 provides a detailed statistical overview of the OmniCharacter++ dataset, encompassing both dyadic and multi-character dialogues in the training and test splits. The training set consists of 88,474 dyadic conversations (average



10.00 turns) and 29,543 multi-character conversations (average 15.05 turns), totaling 118,017 dialogue samples and covering 10,277 unique characters. The test set contains 185 dyadic conversations (average 9.89 turns) and 334 multi-character conversations (average 16.72 turns), featuring 10 held-out characters. The dataset also provides extensive speech annotations, including 2867.94 hours of audio for dyadic dialogues and 1051.66 hours for multi-character dialogues in the training set, as well as 6.96 and 15.20 hours in the test set, respectively. As shown in Fig. 3, OmniCharacter++ widely spans 8 real-world topics, including Negotiation, Exchange, Free-talk, Expert-domain, Instruction-giving, Persuasion, Conflict-resolution, and Planning. Each topic is further decomposed into specific tags such as collaboration, daily life, and cultural exchange, yielding a total of 31 fine-grained topic categories. In addition, Fig. 4 shows the distribution of dialogue turns (left) and audio duration (right) across the dataset. On one hand, most dialogues are longer than 6 turns, highlighting the dataset’s focus on multi-turn interactions. On the other hand, the audio duration statistics further reveal that the speech in the dataset is sufficiently long to enable vivid expressions. These statistics emphasize the dataset’s capability to support the evaluation of RPAs in facilitating multi-modal conversations.

To examine potential templated artifacts, we analyze dialogue diversity in OmniCharacter++ and compare it with CharacterEval [14]. Using standard metrics (Distinct-1/2 [59], Self-BLEU [60]), results in Tab. 3 show higher diversity and lower repetition, indicating more natural, non-templated dialogues. We further analyze character distributions across persona categories and personality traits, as shown in Fig. 5 and Fig. 6. As training and test sets share the same distribution, we report training statistics. Results show balanced coverage and diverse, evenly distributed traits, ensuring both structural balance and diversity.

In this section, we elaborate on our proposed UniCharacter, a novel baseline model capable of simultaneously perceiv-

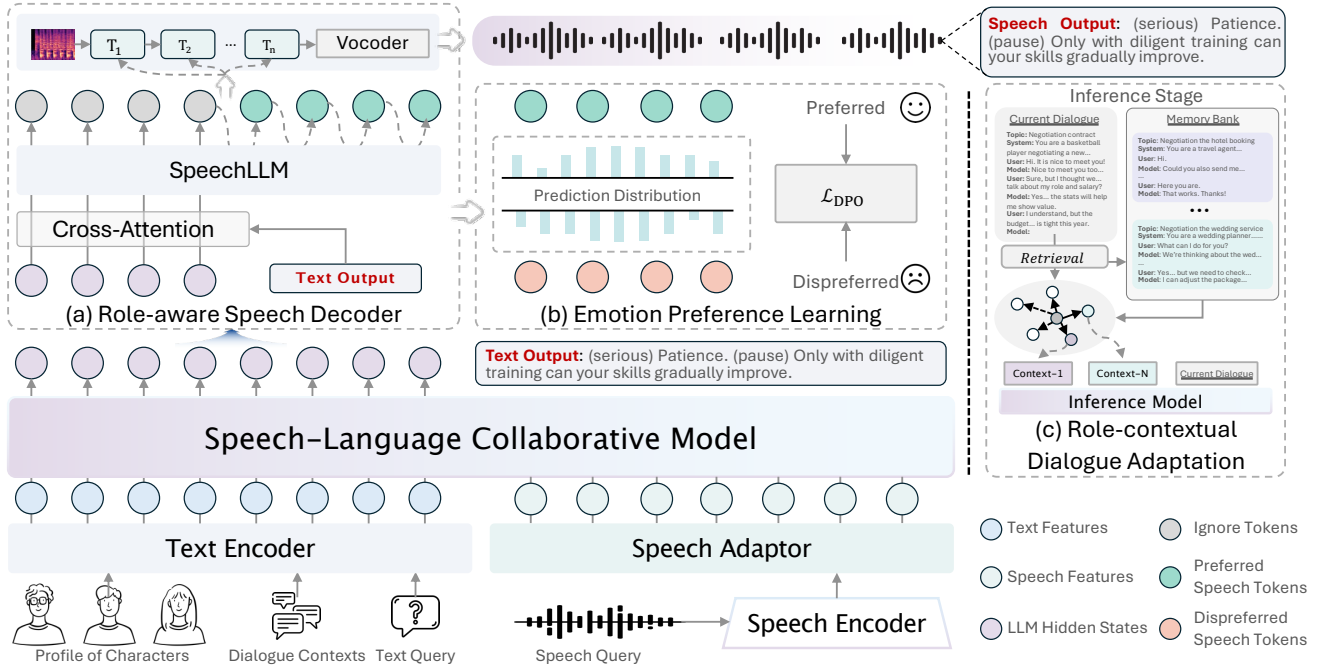


Fig. 7: The overview of UniCharacter framework. We first build a speech-language collaborative model, a large language model that receives both speech and language inputs for unified modeling. Further, a role-aware speech decoder takes the hidden state of the large language model and speech tokens as input to synthesize character-related speech responses. Moreover, an emotion preference learning is devised to determine the suitable emotional expression of speech response, based on the current dialogue scenario. Finally, we employ a role-contextual dialogue adaptation to improve the model's grasp of context at inference by including relevant in-context examples aligned with the ongoing discussion.

ing both speech and language while generating character-specific audio-text responses with superior in-context adaptation ability. The overall framework is shown in Fig. 7.

4.1 Overview

The traditional RPAs are designed to make LLMs interact with users or other characters by emulating a specific character. To achieve this, RPAs utilize a profile denoted as P , along with the current dialogue contexts $C_{n-1} = [(q_i, r_i)]_{i=1}^{n-1}$ to generate a response r_n consistently with character profile:

$$r_n = \text{RPAs}(q_n | C_{n-1}, P), \quad (1)$$

where q_i represent the i -th input query and r_i indicates the corresponding response. However, this paradigm primarily relies on the behavior cloning capabilities of LLMs and places a strong emphasis on producing textual responses. While effective for certain scripted scenarios, it falls short in capturing the richer dynamics of human communication, where interaction often involves engaging with multiple interlocutors through multi-modal ways.

To bridge this gap, we propose UniCharacter, which models persona by involving speech and language modalities for role-playing, enabling controllable emotional speech synthesis with context-aware dialogues. Specifically, UniCharacter first establishes a unified **Speech-Language Collaborative Model**, primarily to achieve semantic alignment by processing textual inputs (individual/multi-character profiles, dialogue contexts, and textual queries) and speech inputs jointly. Building upon this foundation, we

further propose three key contributions to achieve realistic RPAs: (1) a **Role-aware Speech Decoder**, designed to generate role-specific speech responses that accurately reflect unique character traits, (2) an **Emotion Preference Learning** module, which adaptively captures and incorporates characters' sentiment preferences to ensure emotion consistency throughout the dialogue, and (3) a **Role-Contextual Dialogue Adaptation**, which dynamically leverages referred dialogue context at inference time to effectively manage and distinguish multiple interacting characters without additional fine-tuning.

Formally, assume we have a conversation between M characters with profiles $P = \{P_m\}_{m=1}^M$, the dialogue context C_{n-1} , as defined in Eq. (1), can be reformulated as $U_{n-1} = [(X_{i,m}, Y_{i,m})]_{i=1}^{n-1}$, where each input query $X_{i,m} \in \{X_{i,m}^S, X_{i,m}^T, X_{i,m}^{S+T}\}$ denotes the i -th query from character $m \in \{1, \dots, M\}$. The model supports three input modalities: audio-only ($X_{i,m}^S$), text-only ($X_{i,m}^T$), or multi-modal ($X_{i,m}^{S+T}$), enabling flexible interactions across diverse character roles. Correspondingly, each response $Y_{i,m} \in \{Y_{i,m}^S, Y_{i,m}^T\}$ represents either an audio or text response generated by character m , to guarantee diversity throughout the conversation. We describe the details of the proposed model in the following subsections.

4.2 Speech-Language Collaborative Model

To achieve comprehensive alignment between speech and language semantics, we begin our method by introducing a unified speech-language collaborative model, which con-

sists of three key components: a speech encoder, a speech adaptor, and a large language model (LLM).

Speech Encoder. To tackle speech input from different characters, we adopt a speech encoder to encode the speech utterance $X_{n,m}^S$ from character m at the n -th dialogue turn into a speech feature sequence $M_{n,m}^S = [M_{n,m}^{S,1}, \dots, M_{n,m}^{S,N_f}]$, where N_f denotes the number of audio frames. It enables fine-grained modeling of speech characteristics for each character, ensuring role-aware and temporally aligned audio representations within the multi-character conversation.

Speech Adaptor. To address the high temporal redundancy of the encoded speech sequence described above, we further apply a down-sampling strategy by grouping every k consecutive frames of each character's speech into a compact sequence. Specifically, for character m at the n -th dialogue turn, its down-sampled speech representation is formulated as $Z_{n,m}^S = [Z_{n,m}^{S,1}, \dots, Z_{n,m}^{S,\lfloor N_f/k \rfloor}]$ for efficient temporal compression while preserving the essential acoustic features at character-level:

$$Z_{n,m}^{S,j} = M_{n,m}^{S,(k*j)} \oplus M_{n,m}^{S,(k*j+1)} \oplus \dots \oplus M_{n,m}^{S,(k*j+k-1)}. \quad (2)$$

This sequence is then encoded by a speech adaptor τ to make it compatible with the embedding space of LLM:

$$E_{n,m}^S = \tau(Z_{n,m}^S). \quad (3)$$

Large Language Model. We utilize Qwen2.5-7B-Instruct [61] as the base LLM, a state-of-the-art model renowned for its advanced capabilities and performance across various benchmarks. For each character m at dialogue turn n , we process the text inputs—including the role profile P_m , dialogue context U_{n-1} , and current text query $X_{n,m}^T$ to construct a tokenized sequence $E_{n,m}^T = \{E_{(n,m),h}^T\}_{h=1}^{N_t}$ via a text encoder, where N_t denote the lengths of the text sequence. It is then concatenated with the corresponding speech representation $E_{n,m}^S = \{E_{(n,m),h}^S\}_{h=1}^{N_s}$, forming a joint input which is then fed into the LLM. Here, N_s indicates the length of the speech sequence. Finally, the LLM is trained in an auto-regressive manner to predict the next text token conditioned on the preceding multi-modal context, ensuring coherent and role-consistent responses across characters:

$$\mathcal{L}_{language} = - \sum_{h=1}^{N_t} \log P(O_h^T | O_{<h}^T), \quad (4)$$

where O_h^T represents the h -th text token from the LLM.

4.3 Role-aware Speech Decoder

Role-context Guided Speech Token Prediction. Intuitively, it is straightforward to use the LLM to predict both text tokens O_h^T and speech tokens O_h^S . However, we observe that this approach leads to training instability during experiment, often resulting in repeated speech token outputs and semantic mismatches between the generated speech and text content. To facilitate stable and accurate speech token prediction, we propose a novel module called Role-Context Guided Speech Token Prediction, built on top of a speech-language collaborative framework. Specifically, it employs a lightweight speech large language model, referred to as SpeechLLM, which leverages the context representations

Algorithm 1: Training Procedure of UniCharacter

Input: Current dialogue contexts U_{n-1} , character profile $P = \{P_m\}_{m=1}^M$, input query $X_{n,m}$, Steps η .

Output: Text-speech responses of character $Y_{n,m}$

Model: Speech Encoder $\mathcal{E}$, Speech Adapter τ , LLM $\mathcal{M}$, SpeechLLM $\mathcal{S}$, Vocoder $\mathcal{V}$.

```

1 # Stage-1: Speech-Language Collaborative Training
2 while not reach  $\eta$  do
3    $E^{P+U} \leftarrow \text{Encode}(P, U_{n-1})$ ;
4   ▷ Encode text query if provided
5    $E^T \leftarrow \text{Encode}(X_{n,m}^T)$ ;
6   ▷ Encode speech query if provided
7    $E^S \leftarrow \tau(\tilde{E}^S)$ ;  $\tilde{E}^S \leftarrow \mathcal{E}(X_{n,m}^S)$ ;
8   #  $Z \leftarrow [E^{P+U}, E^T]$ ; // Text
9   #  $Z \leftarrow [E^{P+U}, E^S]$ ; // Speech
10   $Z \leftarrow [E^{P+U}, E^T, E^S]$ ; // multi-modal
11   $O^T = \mathcal{M}(Z)$ ;
12  Update  $\tau$  and  $\mathcal{M}$  by  $\mathcal{L}_{language}$ .

13 # Stage-2: Character Speech Token Generation
14 while not reach  $\eta$  do
15   ▷ P-1: Role-context Guided Speech Token Gen.
16    $O^S = \mathcal{S}(H')$ ;  $H' = \text{CrossAtt}(H, O^T)$ ;
17   Convert  $O^S$  to waveform  $Y_{n,m}^S = \mathcal{V}(O^S)$ ;
18   ▷ P-2: Emotion Preference Learning
19    $y_w \leftarrow O^S$ ; // preferred emotion
20    $y_l \leftarrow \tilde{O}^S = \pi_{ref}(H, O^T)$ ; // repeat P-1
21   Update  $\mathcal{S}$  and CrossAtt using  $\mathcal{L}_{speech}$  and  $\mathcal{L}_{DPO}$ .
```

$H = [H_1, \dots, H_{N_t+N_s}]$ produced by the LLM to effectively generate speech tokens. These context representations encode rich information, including the multi-party dialogue semantics and the current query intent, providing SpeechLLM with the necessary cues for coherent and contextually appropriate speech generation.

$$O^S = \text{SpeechLLM}(O_h^S | O_{1:h-1}^S, \phi(H)). \quad (5)$$

Here, ϕ denotes a linear projection layer that maps the context representations into the representation space of SpeechLLM. This design not only stabilizes training but also ensures that the generated speech tokens are well aligned with the contextual semantics, effectively mitigating the issue of repetitive speech token outputs. Importantly, SpeechLLM is also optimized using an auto-regressive training objective:

$$\mathcal{L}_{speech} = - \sum_{h=1}^{N_s} \log P(O_h^S | O_{<h}^S). \quad (6)$$

Role-aware Speech Synthesis. A high-quality speech response that captures each character's traits and persona is essential for building realistic multi-character interactions. To this end, we further introduce a Role-Aware Speech Synthesis module, which generates character-specific audio responses from predicted speech tokens. This module ensures that each character in the dialogue maintains a consistent and distinguishable identity, enhancing the realism of multi-party interactions. Following the approach of [40],

we first decode the speech tokens into a Mel spectrogram, which is then used to synthesize the final waveform. To achieve that, we adopt a Conditional Flow Matching (CFM) model that conditions the sampling process on the context representations H and the speech tokens O^S . The sampling process can be defined by a time-dependent vector field $T = \{T_1, \dots, T_n\}$. To improve efficiency, the optimal-transport (OT) flow is used. We summarize the above process as:

$$Y_{n,m}^S = \text{OT-CFM}(p_0|H, O^S), \quad (7)$$

where p_0 is the initial Mel spectrogram. Ultimately, after obtaining the generated Mel spectrogram, a HiFi-GAN vocoder serves to synthesize a continuous waveform from the Mel spectrogram, facilitating the conversion of spectral representations into high-fidelity audio signals $Y_{n,m}^S$ containing characters' traits such as voice style.

4.4 Emotion Preference Learning

Although our model can generate audio responses that are consistent with the dialogue context, the emotional expressiveness of these responses remains limited. This is primarily because speech generation is conditioned on implicit textual features, which often fail to capture the strong acoustic dynamics. As a result, the generated speech may sound flat, reducing its emotion and weakening the character's personality. To address this limitation, UniCharacter introduces an Emotion Preference Learning strategy that guides the model toward producing emotionally rich and contextually aligned speech.

To solve this, the model requires explicit supervision signals that distinguish contextually appropriate speech from inappropriate emotional variants. Therefore, we construct preference pairs for training. We generate negative samples from the original speech $Y_{n,m}^S$ by modifying its speaking style to an opposite or neutral emotion. It is achieved by using the Speaking Style Control mechanism of CosyVoice 2 [62], which allows explicit manipulation of speaking styles of audio such as joy, anger, sadness, etc. The original $Y_{n,m}^S$ with the target emotion is treated as the positive sample (y_w), while its emotion-altered counterpart serves as the negative sample (y_l). Then, the training objective is to maximize the likelihood of generating emotionally congruent speech sequences while minimizing the preference for emotionally inconsistent ones. We adopt a Direct Preference Optimization (DPO) objective based on contrastive training between the preference pairs. The training loss is defined as:

$$\mathcal{L}_{DPO} = -\log \sigma \left(\beta \cdot \left[\log \frac{\pi^*(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi^*(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right] \right), \quad (8)$$

where β is a temperature scaling factor, σ is the sigmoid function, and π^* is the policy model being optimized. The reference model π_{ref} is used as a stable baseline during training. Note that although preference optimization has been explored in audio-language modeling (e.g., Qwen2-Audio [63], EMOVA [64], and AnyGPT [65]), existing approaches are largely limited to single-turn settings or rely on explicit emotion supervision (e.g., predefined labels), restricting their ability to model context-dependent emotional dynamics. In contrast, our approach targets multi-turn,

Algorithm 2: Inference Procedure of UniCharacter

Input: Current dialogue context U_{n-1} , dialogue memory $\mathcal{D} = \{D_1, \dots, D_L\}$, number of in-context examples k

Output: Generated character response $Y_{n,m}$

Model: LDA topic extractor $\mathcal{T}$, Jensen-Shannon Divergence $\text{JSD}(\cdot, \cdot)$, LLM $\mathcal{M}$

```

1 # Step-1: Topic Extraction
2  $z_C \leftarrow \mathcal{T}(U_{n-1});$  // current topic
3 for  $i = 1$  to  $L$  do
4    $z_{D_i} \leftarrow \mathcal{T}(D_i);$  // memorized topics

5 # Step-2: Topic Similarity Ranking
6 for  $i = 1$  to  $L$  do
7    $s_i \leftarrow \text{JSD}(z_C, z_{D_i})$ 
8 Retrieve top- $k$  dialogues with lowest  $s_i$  values;
9  $\mathcal{E} = \{E_1, \dots, E_k\};$  // In-context examples

10 # Step-3: Construct Input and Inference
11  $Y_{n,m} \leftarrow \text{Infer}([E_1, \dots, E_k, P, U_{n-1}, X_{n,m}])$ 

```

multi-character role-playing scenarios, where emotional expression must remain consistent with both dialogue context and character identity. By formulating emotion alignment as a context-aware preference learning problem, we enable implicit modeling of emotional transitions across turns, resulting in more coherent and natural interactions.

4.5 Role-Contextual Dialogue Adaptation

The motivation of the Role-Contextual Dialogue Adaptation is that humans naturally rely on contextual memory to interpret and respond appropriately in an ongoing dialogue. Inspired by this, we proposed such a strategy to enhance the model's context understanding by incorporating in-context examples having relevant topics with the current conversation. To construct these examples, we adopt a retrieval-based approach. Given a current dialogue context U_{n-1} and a memory containing L candidate dialogues $\mathcal{D} = \{D_1, \dots, D_L\}$, we first extract their associated topic distributions using Latent Dirichlet Allocation (LDA) $\mathcal{T}(\cdot)$:

$$z_C = \mathcal{T}(C_n), \quad z_{D_i} = \mathcal{T}(D_i), \quad i = 1, \dots, L \quad (9)$$

We then compute topic-level similarity between the current dialogue and each candidate using the Jensen-Shannon divergence (JSD) [66], which is a symmetric and smoothed version of the Kullback-Leibler divergence. Given two topic distributions Z_c and Z_{D_i} (denoted as G and Q), the JSD is defined as:

$$\text{JSD}(G \parallel Q) = \frac{1}{2} \text{KL}(G \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad (10)$$

where $M = \frac{1}{2}(G + Q)$

Based on the JSD scores, we retrieve the top- k candidate dialogues with the most similar topic distributions to z_C . These top- k dialogues are used as in-context examples, denoted as $\mathcal{E} = \{E_1, \dots, E_k\}$, and prepended to the input during inference:

$$\text{Input}_{\text{inference}} = [E_1, \dots, E_k, *], \quad (11)$$

where $*$ denotes the remaining input components, including multi-role profiles, dialogue history, current query, and other task-specific context. By following the natural paradigm of human conversation, this strategy guides the model towards generating responses that are coherent and contextually grounded without any additional training or parameter updates.

4.6 Training Strategy

Following [37], we adopt a two-stage training strategy for UniCharacter. In the first stage, we train the speech adapter and LLM to generate text responses based on the text and speech inputs using the objective $\mathcal{L}_{\text{language}}$ in Eq. (4). In the second stage, we only train the cross-attention and SpeechLLM for speech token prediction by using the objective $\mathcal{L}_{\text{speech}}$ in Eq. (6). Additionally, we apply the emotion preference learning objective $\mathcal{L}_{\text{DPO}}$ in Eq. (8) to enhance emotional expressiveness during this stage jointly. For role-aware speech synthesis, we utilize the pre-trained conditional flow matching and HiFi-GAN [67] weights from GLM-4-Voice [40] and fine-tune them on our data to obtain role-specific voice generation. The overall training and inference procedures of UniCharacter are illustrated in **Algorithms 1 and 2**.

5 EXPERIMENTS

In this section, we conduct extensive experiments on the OmniCharacter++ benchmark to evaluate the role-playing capabilities of state-of-the-art models and the proposed UniCharacter. We first describe detailed experimental setup in Section 5.1. Next, we perform comprehensive evaluations across a wide range of models, including proprietary models, open-source models, reasoning models, and role-playing models in Section 5.2. We then provide an in-depth analysis of UniCharacter's key components in Section 5.3. At last, we compare the role-playing performance of UniCharacter by training it on OmniCharacter++ while evaluating it on the challenging OOD role-playing benchmark CharacterEval [14] to validate its generalization in Section 5.4.

5.1 Experimental Setup

Benchmark Models. The evaluated models on OmniCharacter++ can be classified into three major categories: **(a) Proprietary Models:** These include GPT-4.1 series [70], GPT-4o series [69], GPT-3.5-Turbo [68], DeepSeek-V3 [71], Doubao-1.5-Pro [72], and Gemini-2.0-flash [73], which are state-of-the-art models accessed via an API under zero-shot settings with standardized prompts. **(b) Open-source Models:** including LLaMA-3.1 series [37], Qwen2.5 Series [81]. All models are deployed locally and use standardized prompts. **(c) Reasoning Models:** We categorize models that internally generate extended chain-of-thought (CoT) reasoning traces as reasoning-oriented models. Representative models in this category contain o1 [77], o3 [76], o4 [76] series, and Gemini-2.5 series [73]. **(d) Role-Playing Models:** We include a distinct category of RPAs models that are tailored for the role-playing task, including CharacterGLM [17], Baichuan-NPC [78], Minimax [79], and Xingchen [80]. **(e) Text-Speech Models:** These models take text/audio as input and generate

text, audio, or both as output, which are designed for tasks like multi-modal dialogue, human-machine interaction, including SpeechGPT [39], SpeechGPT2 [41], LLaMA-Omni [37], OpenOmni [82], Qwen2.5-Omni [83], and our previous OmniCharacter [29].

Evaluation Metrics. We evaluate role-playing capabilities from two perspectives: dyadic and multi-party dialogue, each under (1) text-level and (2) acoustic-level evaluation. **Text-level evaluation** includes: (1) *Context Understanding*, which measures discriminative ability via multiple-choice questions given role definitions and dialogue context. Following MMBench [36], we adopt a Circular Evaluation Strategy, where options are cyclically shifted across N inference passes, and a question is deemed correct only if all predictions are correct, yielding a more stringent assessment. (2) *Conversation Ability*, which evaluates response quality given role profiles and context. Outputs are assessed by multiple LLM judges (GPT-4o, Gemini-1.5-Pro, Claude-3.5-Sonnet) across Topic Following (TF), Goal Success (GS), Character Consistency (CC), and Dialogue Coherence (DC). **Acoustic-level evaluation** focuses on *Human Perception*, where five experts assess model outputs across Fluency, Consistency, Emotional Expression, Clarity, Appropriateness, and Immersion. All evaluations are conducted on the OmniCharacter++ test set, with each expert covering all samples.

Besides, to further assess the generalization ability of our model, we also evaluate UniCharacter on CharacterEval [14], the most widely used benchmark for role-playing agents (RPAs) to date. Note that the roles in OmniCharacter++ test set and CharacterEval are all out-of-distribution (OOD) roles, *i.e.*, without overlap with those in OmniCharacter++ train set.

Implementation Details. We employ the Whisper-large-v3 [57] as the speech encoder and Qwen-2.5-7B-Instruct [81] as the LLM in speech-language collaborative model. For role-context guided speech token prediction, we employ the lightweight Qwen2.5-0.5B-Instruct [81] model as SpeechLLM to generate speech tokens. The weights of the speech tokenizer and role-aware speech synthesis are initialized from the pretrained checkpoint of GLM-4-Voice [40]. The speech-language collaborative model and role-aware speech decoder are initialized using pre-trained weights from OpenOmni [82], and we conduct model fine-tuning on the OmniCharacter++ train set to obtain our UniCharacter. All speech data are resampled to a frequency of 16Khz.

Training Details The UniCharacter employs a two-stage training strategy as described in Section 4.6. In the first stage, we utilize the AdamW optimizer [84] with a warmup ratio of 0.03, gradually increasing the learning rate until it reaches $2e-7$. The second stage retains the parameters established in the first stage and adjusts only the final learning rate to $2e-8$. We set the batch size to 1 for both stages. The training of each stage is finished within 1 epoch on $8 \times A6000$ GPUs.

5.2 Comparison with State of the Arts

To certify the ability of various models for realistic role-playing, we give a quantitative analysis based on OmniCharacter++ benchmark across three main evaluation dimensions, *i.e.*, context understanding, conversation ability, and human performance.

TABLE 4: **Performance comparison with state-of-the-art models on OmniCharacter++ test set of dyadic dialogue (Context Understanding).** The results are evaluated on the dyadic and multi-character conversation scenarios, under the setting of multi-choice QA. All models are tested based on the Circular Evaluation Strategy [36] for a more robust evaluation, alleviating the negative impact of noise and random prediction of models. Neg.: negotiation, Exc.: exchange, Free.: free-talk, Exp.: expert-domain, Inst.: instruction-giving, Per.: persuasion, Conf.: conflict-resolution, Pla: planning. The number in () indicates the ranks.

Models	Avg.	Dyadic Dialogue - Context Understanding (Multi-Choice)							
		Neg.	Exc.	Free.	Exp.	Inst.	Per.	Conf.	Pla.
Human Evaluation									
Human	93.61 (-)	92.81 $\pm$ 1.0	94.05 $\pm$ 1.0	94.58 $\pm$ 1.2	96.47 $\pm$ 1.1	93.68 $\pm$ 1.0	91.74 $\pm$ 1.0	89.92 $\pm$ 0.9	95.63 $\pm$ 1.3
Blind Evaluation (<i>w/o</i> dialogue context)									
Random Choice	26.66 (-)	33.33	16.67	33.33	25.00	14.29	25.00	11.11	54.55
Random Choice (circular eval.)	0.00 (-)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
GPT-3.5-Turbo [68]	28.21 (-)	24.87 $\pm$ 1.1	30.46 $\pm$ 1.4	32.41 $\pm$ 1.0	25.33 $\pm$ 0.9	33.09 $\pm$ 1.1	33.88 $\pm$ 1.1	18.72 $\pm$ 1.1	26.94 $\pm$ 1.1
GPT-4o [69]	28.26 (-)	24.92 $\pm$ 1.0	34.11 $\pm$ 1.1	34.02 $\pm$ 0.9	19.68 $\pm$ 0.8	27.95 $\pm$ 1.0	45.87 $\pm$ 1.1	20.66 $\pm$ 0.8	18.88 $\pm$ 1.1
Proprietary Models									
GPT-4.1 [70]	57.55 (1)	44.32 $\pm$ 1.1	62.11 $\pm$ 1.2	77.02 $\pm$ 1.3	75.12 $\pm$ 1.3	49.28 $\pm$ 1.0	54.77 $\pm$ 1.1	48.12 $\pm$ 1.2	49.66 $\pm$ 0.9
GPT-4.1-mini [70]	47.40 (5)	36.11 $\pm$ 1.2	44.88 $\pm$ 1.2	64.32 $\pm$ 1.1	57.88 $\pm$ 1.0	47.21 $\pm$ 1.2	40.66 $\pm$ 1.1	41.12 $\pm$ 1.2	47.02 $\pm$ 1.1
GPT-4o [69]	52.27 (4)	46.02 $\pm$ 1.1	59.88 $\pm$ 1.1	74.66 $\pm$ 1.0	58.11 $\pm$ 1.2	38.77 $\pm$ 1.2	50.62 $\pm$ 0.9	45.11 $\pm$ 0.9	45.02 $\pm$ 1.0
GPT-4o-mini [69]	39.13 (7)	28.77 $\pm$ 1.1	40.66 $\pm$ 1.1	56.88 $\pm$ 1.0	53.44 $\pm$ 1.2	36.88 $\pm$ 1.2	38.77 $\pm$ 1.0	30.88 $\pm$ 1.1	26.77 $\pm$ 1.0
GPT-3.5-Turbo [68]	29.94 (8)	30.77 $\pm$ 1.2	25.88 $\pm$ 1.1	34.66 $\pm$ 0.9	29.77 $\pm$ 0.7	26.44 $\pm$ 0.9	30.88 $\pm$ 0.9	31.22 $\pm$ 1.1	29.88 $\pm$ 1.0
DeepSeek-V3 [71]	46.84 (6)	40.77 $\pm$ 0.7	50.88 $\pm$ 1.1	64.44 $\pm$ 0.9	53.22 $\pm$ 1.1	45.88 $\pm$ 1.0	43.02 $\pm$ 0.9	35.66 $\pm$ 1.1	40.88 $\pm$ 1.2
Doubao-1.5-Pro-32K [72]	54.92 (3)	44.88 $\pm$ 0.7	55.66 $\pm$ 1.1	57.77 $\pm$ 1.0	60.88 $\pm$ 1.1	63.11 $\pm$ 1.0	52.88 $\pm$ 1.2	50.44 $\pm$ 1.1	53.77 $\pm$ 0.8
Gemini-2.0-flash-preview [73]	55.84 (2)	50.11 $\pm$ 1.1	59.88 $\pm$ 1.3	74.66 $\pm$ 1.1	53.44 $\pm$ 0.8	55.88 $\pm$ 0.9	55.11 $\pm$ 1.1	48.77 $\pm$ 1.0	48.88 $\pm$ 0.8
Open-source Models									
LLaMA-3.1-405B-Instruct [74]	46.86 (2)	41.77 $\pm$ 1.3	45.88 $\pm$ 1.1	46.02 $\pm$ 1.1	49.66 $\pm$ 1.2	45.77 $\pm$ 0.8	48.11 $\pm$ 1.0	47.88 $\pm$ 1.1	49.77 $\pm$ 1.1
LLaMA-3.1-70B-Instruct [74]	43.46 (5)	41.55 $\pm$ 1.0	46.77 $\pm$ 1.0	42.88 $\pm$ 1.0	41.77 $\pm$ 1.1	46.12 $\pm$ 1.2	42.66 $\pm$ 1.2	46.02 $\pm$ 1.1	39.88 $\pm$ 1.3
LLaMA-3.1-8B-Instruct [74]	30.13 (7)	31.88 $\pm$ 1.1	26.11 $\pm$ 0.9	27.66 $\pm$ 1.1	19.88 $\pm$ 1.2	39.11 $\pm$ 1.3	32.55 $\pm$ 1.4	34.77 $\pm$ 0.9	29.11 $\pm$ 1.1
Qwen2.5-72B-Instruct [75]	50.87 (1)	42.55 $\pm$ 1.0	43.11 $\pm$ 0.9	62.88 $\pm$ 0.7	66.77 $\pm$ 1.2	55.44 $\pm$ 1.1	43.77 $\pm$ 1.1	48.88 $\pm$ 1.2	43.55 $\pm$ 1.3
Qwen2.5-32B-Instruct [75]	46.04 (3)	37.88 $\pm$ 1.0	46.11 $\pm$ 0.8	62.02 $\pm$ 1.1	55.88 $\pm$ 1.0	43.77 $\pm$ 1.1	52.88 $\pm$ 1.2	34.11 $\pm$ 1.2	35.66 $\pm$ 1.1
Qwen2.5-14B-Instruct [75]	43.91 (4)	32.88 $\pm$ 1.2	33.44 $\pm$ 1.3	64.88 $\pm$ 1.0	60.02 $\pm$ 0.9	41.77 $\pm$ 1.2	37.55 $\pm$ 1.2	40.66 $\pm$ 1.3	40.11 $\pm$ 1.1
Qwen2.5-7B-Instruct [75]	30.66 (6)	26.44 $\pm$ 1.0	29.11 $\pm$ 0.7	42.77 $\pm$ 1.2	40.11 $\pm$ 1.0	20.88 $\pm$ 1.2	34.55 $\pm$ 1.1	28.66 $\pm$ 1.3	22.77 $\pm$ 1.2
Reasoning Models									
o4-mini [76]	46.39 (4)	41.77 $\pm$ 1.0	48.11 $\pm$ 1.1	57.02 $\pm$ 1.1	55.66 $\pm$ 1.1	43.88 $\pm$ 1.1	47.22 $\pm$ 0.9	33.77 $\pm$ 1.0	43.66 $\pm$ 1.2
o3-mini [76]	49.11 (3)	44.55 $\pm$ 1.2	52.11 $\pm$ 1.2	62.88 $\pm$ 1.4	55.88 $\pm$ 1.2	43.77 $\pm$ 1.0	47.88 $\pm$ 1.3	38.11 $\pm$ 1.1	47.66 $\pm$ 1.5
o1-mini [77]	43.72 (5)	37.88 $\pm$ 1.1	43.77 $\pm$ 1.1	50.88 $\pm$ 1.0	55.88 $\pm$ 1.0	43.77 $\pm$ 0.9	45.11 $\pm$ 1.1	36.77 $\pm$ 1.1	35.66 $\pm$ 1.4
Gemini-2.5-flash [73]	49.48 (2)	41.55 $\pm$ 1.0	41.22 $\pm$ 0.9	50.88 $\pm$ 1.2	56.88 $\pm$ 1.2	44.55 $\pm$ 1.2	52.11 $\pm$ 1.3	53.77 $\pm$ 1.3	54.88 $\pm$ 0.8
Gemini-2.5-pro-preview-05-06 [73]	52.99 (1)	47.88 $\pm$ 0.8	47.55 $\pm$ 1.1	60.22 $\pm$ 1.2	61.88 $\pm$ 1.2	48.55 $\pm$ 1.3	52.44 $\pm$ 1.1	53.88 $\pm$ 1.1	51.55 $\pm$ 1.0
Role-playing Models									
CharacterGLM [17]	43.46 (6)	41.55 $\pm$ 1.0	41.77 $\pm$ 1.1	68.12 $\pm$ 1.1	49.77 $\pm$ 0.9	27.22 $\pm$ 0.9	46.11 $\pm$ 0.8	39.88 $\pm$ 1.3	33.22 $\pm$ 1.1
Baichuan-NPC [78]	43.27 (7)	36.88 $\pm$ 1.3	46.11 $\pm$ 1.3	56.02 $\pm$ 1.2	49.77 $\pm$ 1.2	38.88 $\pm$ 1.2	37.88 $\pm$ 1.2	39.77 $\pm$ 1.1	40.88 $\pm$ 0.9
Minimax-abab6-chat [79]	49.24 (4)	49.11 $\pm$ 1.1	46.55 $\pm$ 1.3	47.88 $\pm$ 1.2	71.55 $\pm$ 1.0	48.77 $\pm$ 1.2	43.88 $\pm$ 1.0	43.55 $\pm$ 1.1	42.66 $\pm$ 0.7
Xingchen-Plus [80]	48.59 (5)	49.88 $\pm$ 1.2	48.77 $\pm$ 1.2	71.66 $\pm$ 1.0	48.55 $\pm$ 1.1	43.66 $\pm$ 1.2	43.55 $\pm$ 1.0	39.77 $\pm$ 0.9	42.88 $\pm$ 0.9
Qwen2.5-7B-Instruct w/ our data	49.70 (3)	46.11 $\pm$ 1.1	53.44 $\pm$ 1.0	74.88 $\pm$ 1.2	43.77 $\pm$ 1.1	50.88 $\pm$ 0.7	43.77 $\pm$ 1.2	43.55 $\pm$ 1.2	41.22 $\pm$ 1.2
OmniCharacter-7B [29] (Ours)	50.23 (2)	41.55 $\pm$ 1.1	55.77 $\pm$ 1.1	73.88 $\pm$ 1.1	41.77 $\pm$ 1.1	42.66 $\pm$ 1.2	43.77 $\pm$ 1.0	53.77 $\pm$ 1.2	48.66 $\pm$ 0.9
UniCharacter-7B (Ours)	55.12 (1)	50.88 $\pm$ 1.1	53.22 $\pm$ 0.9	78.11 $\pm$ 0.9	48.77 $\pm$ 1.1	52.11 $\pm$ 1.1	51.22 $\pm$ 1.2	58.77 $\pm$ 1.0	47.88 $\pm$ 1.2

5.2.1 Context Understanding Performance

To further evaluate contextual understanding under varying interaction complexity, we split the OmniCharacter++ test set into dyadic and multi-party dialogue scenarios and report results in Tab. 4 and Tab. 5, respectively. The key findings are as follows. **i) Multi-party dialogue is substantially more challenging.** All models show consistent degradation when moving from dyadic to multi-party settings, indicating that multi-speaker interaction introduces additional complexity in speaker tracking and role dependency modeling. **ii) Performance hierarchy remains but robustness differs.** Proprietary models (e.g., GPT-4.1, Gemini-2.0-flash) achieve the best results overall, while large open-source models remain competitive in dyadic dialogue but degrade more noticeably in multi-party settings, suggesting weaker robustness to complex interaction structures. **iii) Reasoning models show limited advantage.** Despite their strength in structured reasoning, reasoning-oriented models do not outperform general LLMs, implying that rigid reasoning patterns are less effective for dynamic, role-dependent dialogue understanding. **iv) UniCharacter achieves strong performance with better generalization.**

UniCharacter ranks first among role-playing models in both settings (55.12%/47.80%), remains competitive with top proprietary models, and exhibits smaller degradation from dyadic to multi-party dialogue, demonstrating stronger robustness in modeling complex role interactions. **v) Our data significantly improves contextual understanding.** Fine-tuning Qwen2.5-7B with our dataset yields substantial gains, highlighting the effectiveness of our data.

5.2.2 Conversation Ability Performance

To comprehensively evaluate the conversation ability for role-playing competence, we further conduct a multi-dimensional evaluation combining both fine-grained scoring metrics and binary task success indicators using three LLM judges: GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonne. In detail, each model is assessed across four key criteria: *Topic Following*: measures the model's ability to remain on-topic relative to the user's query or the ongoing dialogue context. *Goal Success*: a binary indicator reflecting whether the model correctly fulfills the instruction or completes the intended dialogue goal. *Character Consistency*: evaluates the internal coherence and stylistic alignment of

TABLE 5: Performance comparison with state-of-the-art models on OmniCharacter++ test set of multi-party dialogue (Context Understanding). The results are evaluated on the dyadic and multi-character conversation scenarios, under the setting of multi-choice QA. All models are tested based on the Circular Evaluation Strategy [36] for a more robust evaluation, alleviating the negative impact of noise and random prediction of models. Neg.: negotiation, Exc.: exchange, Free.: free-talk, Exp.: expert-domain, Inst.: instruction-giving, Per.: persuasion, Conf.: conflict-resolution, Pla: planning. The number in () indicates the ranks.

Models	Avg.	Multi-party Dialogue - Context Understanding (Multi-Choice)							
		Neg.	Exc.	Free.	Exp.	Inst.	Per.	Conf.	Pla.
Human Evaluation									
Human	89.84 (-)	88.66 $\pm$ 1.0	90.88 $\pm$ 1.1	91.11 $\pm$ 1.0	92.77 $\pm$ 1.0	89.88 $\pm$ 1.2	87.44 $\pm$ 1.1	85.88 $\pm$ 0.7	92.11 $\pm$ 1.1
Blind Evaluation (<i>w/o</i> dialogue context)									
Random Choice	23.14 (-)	24.02 $\pm$ 1.1	23.88 $\pm$ 1.2	18.44	27.77 $\pm$ 1.0	22.66 $\pm$ 1.4	20.11 $\pm$ 1.2	24.66 $\pm$ 1.0	23.55 $\pm$ 1.3
Random Choice (circular eval.)	0.00 (-)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
GPT-3.5-Turbo [68]	21.51 (-)	18.44 $\pm$ 1.1	23.77 $\pm$ 1.0	25.88 $\pm$ 1.2	18.55 $\pm$ 1.1	26.11 $\pm$ 1.3	27.66 $\pm$ 1.1	11.22 $\pm$ 1.1	20.44 $\pm$ 1.0
GPT-4o [69]	24.68 (-)	21.11 $\pm$ 0.9	30.02 $\pm$ 1.2	30.44 $\pm$ 0.7	16.22 $\pm$ 1.1	24.88 $\pm$ 1.0	41.77 $\pm$ 1.0	17.11 $\pm$ 1.0	15.88 $\pm$ 1.2
Proprietary Models									
GPT-4.1 [70]	50.11 (1)	37.44 $\pm$ 1.2	54.22 $\pm$ 1.0	69.11 $\pm$ 1.3	67.88 $\pm$ 1.1	42.11 $\pm$ 1.4	46.77 $\pm$ 1.2	41.11 $\pm$ 1.3	42.22 $\pm$ 0.9
GPT-4.1-mini [70]	40.55 (5)	29.44 $\pm$ 1.2	38.11 $\pm$ 1.2	57.44 $\pm$ 1.1	50.88 $\pm$ 1.2	40.11 $\pm$ 1.1	34.11 $\pm$ 1.1	34.22 $\pm$ 1.0	40.11 $\pm$ 1.2
GPT-4o [69]	45.20 (4)	39.11 $\pm$ 1.1	52.22 $\pm$ 1.2	66.88 $\pm$ 1.3	50.88 $\pm$ 1.3	31.77 $\pm$ 1.3	43.77 $\pm$ 1.0	38.77 $\pm$ 1.0	38.22 $\pm$ 1.1
GPT-4o-mini [69]	32.12 (7)	21.11 $\pm$ 1.2	33.77 $\pm$ 1.2	50.11 $\pm$ 1.0	46.77 $\pm$ 1.3	30.11 $\pm$ 1.3	31.88 $\pm$ 1.0	24.11 $\pm$ 1.0	19.11 $\pm$ 1.1
GPT-3.5-Turbo [68]	22.90 (8)	23.88 $\pm$ 1.2	18.44 $\pm$ 1.0	27.11 $\pm$ 0.9	22.88 $\pm$ 0.9	20.11 $\pm$ 1.0	23.88 $\pm$ 1.3	24.11 $\pm$ 1.2	22.77 $\pm$ 1.0
DeepSeek-V3 [71]	39.94 (6)	33.77 $\pm$ 0.9	43.88 $\pm$ 1.3	57.44 $\pm$ 0.6	46.77 $\pm$ 1.1	38.77 $\pm$ 1.0	36.22 $\pm$ 0.9	28.44 $\pm$ 1.2	34.22 $\pm$ 1.1
Doubao-1.5-Pro-32K [72]	47.66 (3)	37.44 $\pm$ 0.4	47.88 $\pm$ 1.1	50.11 $\pm$ 1.0	52.88 $\pm$ 1.2	56.11 $\pm$ 1.1	46.22 $\pm$ 1.3	43.88 $\pm$ 1.2	46.77 $\pm$ 0.9
Gemini-2.0-flash-preview [73]	48.36 (2)	42.77 $\pm$ 1.1	52.22 $\pm$ 1.3	66.88 $\pm$ 1.1	46.77 $\pm$ 0.9	48.88 $\pm$ 1.0	48.11 $\pm$ 1.4	41.11 $\pm$ 1.2	40.11 $\pm$ 0.7
Open-source Models									
LLaMA-3.1-405B-Instruct [74]	39.75 (2)	34.77 $\pm$ 1.3	38.88 $\pm$ 1.4	39.22 $\pm$ 1.2	41.88 $\pm$ 1.2	39.11 $\pm$ 0.9	41.11 $\pm$ 1.3	40.88 $\pm$ 1.1	42.11 $\pm$ 1.0
LLaMA-3.1-70B-Instruct [74]	36.21 (5)	34.77 $\pm$ 1.0	39.11 $\pm$ 1.0	36.11 $\pm$ 1.1	35.11 $\pm$ 1.2	39.11 $\pm$ 1.2	35.22 $\pm$ 1.1	38.11 $\pm$ 1.0	32.11 $\pm$ 1.2
LLaMA-3.1-8B-Instruct [74]	22.94 (7)	25.11 $\pm$ 1.1	19.11 $\pm$ 0.9	20.11 $\pm$ 1.2	12.11 $\pm$ 1.2	31.77 $\pm$ 1.1	25.11 $\pm$ 1.0	28.11 $\pm$ 0.8	22.11 $\pm$ 1.3
Qwen2.5-72B-Instruct [75]	43.59 (1)	34.88 $\pm$ 1.0	36.77 $\pm$ 0.9	54.11 $\pm$ 0.8	59.11 $\pm$ 1.1	49.11 $\pm$ 1.2	36.88 $\pm$ 1.2	41.11 $\pm$ 1.0	36.77 $\pm$ 1.3
Qwen2.5-32B-Instruct [75]	38.49 (3)	30.11 $\pm$ 1.0	39.11 $\pm$ 0.8	54.11 $\pm$ 1.1	48.11 $\pm$ 1.0	36.11 $\pm$ 1.0	45.11 $\pm$ 1.0	27.11 $\pm$ 1.0	28.11 $\pm$ 1.2
Qwen2.5-14B-Instruct [75]	36.91 (4)	25.11 $\pm$ 1.2	26.77 $\pm$ 1.3	57.11 $\pm$ 0.9	52.11 $\pm$ 0.6	35.11 $\pm$ 1.2	30.88 $\pm$ 1.0	34.11 $\pm$ 1.3	34.11 $\pm$ 1.2
Qwen2.5-7B-Instruct [75]	23.33 (6)	19.11 $\pm$ 1.0	21.11 $\pm$ 0.7	35.11 $\pm$ 1.1	32.88 $\pm$ 1.1	14.11 $\pm$ 1.0	27.11 $\pm$ 1.0	21.11 $\pm$ 1.2	16.11 $\pm$ 1.2
Reasoning Models									
o4-mini [76]	38.91 (4)	34.88 $\pm$ 1.1	40.11 $\pm$ 1.1	49.11 $\pm$ 1.2	48.11 $\pm$ 1.0	36.11 $\pm$ 1.1	39.11 $\pm$ 0.8	27.11 $\pm$ 1.0	36.77 $\pm$ 1.0
o3-mini [76]	41.15 (3)	37.44 $\pm$ 1.2	44.11 $\pm$ 1.1	54.11 $\pm$ 1.2	48.11 $\pm$ 1.4	36.11 $\pm$ 1.2	39.11 $\pm$ 1.2	30.11 $\pm$ 1.2	40.11 $\pm$ 1.2
o1-mini [77]	35.82 (5)	30.11 $\pm$ 1.0	36.77 $\pm$ 1.0	42.11 $\pm$ 1.1	48.11 $\pm$ 1.1	36.11 $\pm$ 1.2	37.11 $\pm$ 1.0	29.11 $\pm$ 1.2	27.11 $\pm$ 1.4
Gemini-2.5-flash [73]	42.33 (2)	34.77 $\pm$ 1.0	34.77 $\pm$ 0.7	43.11 $\pm$ 1.2	49.11 $\pm$ 1.1	37.11 $\pm$ 1.2	45.11 $\pm$ 1.2	46.77 $\pm$ 1.3	47.88 $\pm$ 0.8
Gemini-2.5-pro-preview-05-06 [73]	45.62 (1)	40.11 $\pm$ 0.8	40.11 $\pm$ 1.1	52.11 $\pm$ 1.0	54.11 $\pm$ 1.2	41.11 $\pm$ 1.1	45.88 $\pm$ 1.3	46.77 $\pm$ 1.1	44.77 $\pm$ 1.0
Role-playing Models									
CharacterGLM [17]	36.47 (6)	34.77 $\pm$ 1.0	35.11 $\pm$ 1.1	60.88 $\pm$ 1.1	42.77 $\pm$ 0.8	20.11 $\pm$ 0.9	39.11 $\pm$ 0.9	32.88 $\pm$ 1.2	26.11 $\pm$ 1.1
Baichuan-NPC [78]	36.23 (7)	29.88 $\pm$ 1.0	38.77 $\pm$ 1.1	49.11 $\pm$ 1.1	42.77 $\pm$ 1.0	31.77 $\pm$ 1.1	30.88 $\pm$ 1.3	32.88 $\pm$ 1.1	33.77 $\pm$ 0.9
Minimax-abab6-chat [79]	42.26 (4)	41.77 $\pm$ 1.0	39.88 $\pm$ 1.2	40.88 $\pm$ 1.0	64.88 $\pm$ 1.0	41.77 $\pm$ 1.2	36.88 $\pm$ 1.1	36.11 $\pm$ 1.1	35.88 $\pm$ 0.6
Xingchen-Plus [80]	41.62 (5)	42.88 $\pm$ 1.2	41.88 $\pm$ 1.1	64.77 $\pm$ 1.0	41.77 $\pm$ 1.1	36.77 $\pm$ 1.2	36.11 $\pm$ 1.3	32.88 $\pm$ 0.9	35.88 $\pm$ 0.9
Qwen2.5-7B-Instruct w/ our data	42.58 (3)	39.11 $\pm$ 1.1	45.88 $\pm$ 1.0	68.11 $\pm$ 1.0	36.77 $\pm$ 1.4	43.77 $\pm$ 0.9	36.77 $\pm$ 1.2	36.11 $\pm$ 1.4	34.11 $\pm$ 1.3
OmniCharacter-7B [29] (Ours)	43.31 (2)	34.77 $\pm$ 1.1	48.88 $\pm$ 1.1	66.88 $\pm$ 1.3	34.77 $\pm$ 1.0	35.88 $\pm$ 1.3	36.77 $\pm$ 1.2	46.77 $\pm$ 1.2	41.77 $\pm$ 0.9
UniCharacter-7B (Ours)	47.80 (1)	43.88 $\pm$ 1.1	46.77 $\pm$ 0.7	70.11 $\pm$ 1.1	41.77 $\pm$ 1.2	44.77 $\pm$ 1.3	44.11 $\pm$ 1.0	50.88 $\pm$ 1.3	40.11 $\pm$ 1.2

the model's responses with the assigned character's persona. *Dialogue Coherence*: assesses how effectively the model leverages prior conversational history to generate contextually appropriate responses. The results are reported in Tab. 6, where all models are evaluated with three independent judges, *i.e.*, GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonnet. Across all judges, UniCharacter achieves the best performance on all metrics, outperforming strong baselines such as GPT-4 and Gemini-2.0-flash. It attains the highest average scores in both dyadic (2.91/2.86/2.88) and multi-party settings (1.80/1.78/1.79), demonstrating stable, judge-agnostic gains. A consistent performance drop is observed from dyadic to multi-party dialogue across all models (*e.g.*, 2.91 $\rightarrow$ 1.80 under GPT-4o), particularly in goal success and character consistency, highlighting the increased difficulty of multi-speaker interactions. OmniCharacter remains competitive, ranking second in most settings and surpassing models such as GPT-3.5, Doubao-1.5-Pro, and Qwen2.5-7B-Instruct. Building on this, UniCharacter further improves all metrics while exhibiting smaller degradation, indicating stronger robustness in complex scenarios.

Multi-Judge Evaluation Stability. With multiple LLM

judges (GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonnet), we observe that the evaluation scores remain largely consistent across different judges on both dyadic and multi-party dialogues. In particular, UniCharacter consistently achieves the best performance, while OmniCharacter ranks second across most evaluation settings. This stable ranking across independent evaluators indicates that the results are not overly sensitive to the choice of a specific LLM judge.

Prompt Perturbation Analysis. To evaluate the sensitivity of LLM-based evaluation to prompt variations, we select GPT-4o as the judge and evaluate under both dyadic and multi-party dialogue settings. Specifically, we consider four prompt variants, including the original prompt (V-1), synonym substitution (V-2), word order variation (V-3), and noise injection (V-4), as shown in Fig. 8. Note that we keep the evaluation criteria unchanged for controllable comparison. As shown in Tab. 7, the evaluation results remain highly consistent across different prompt variants in both settings, with only minor fluctuations. These results indicate that the evaluation is robust to prompt perturbations and not sensitive to superficial variations in prompt formulation.

Inter-Judge Agreement. We measure the agreement be-

TABLE 6: Performance comparison with state-of-the-art models on OmniCharacter++ test set (Conversation Ability). Scores are reported using three judges: GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonnet. TF: Topic Following (1–5), GS: Goal Success (1 = success, 0 = failure), CC: Character Consistency (1–5), DC: Dialogue Coherence (1–5).

Judge	Models	TF	GS	CC	DC	Avg.
Dyadic Dialogue	GPT-4 [70]	3.51	0.64	3.33	3.49	2.76
	GPT-3.5 [68]	3.18	0.59	3.07	2.96	2.47
	Doubao-1.5-Pro-256K [72]	2.94	0.58	2.82	2.89	2.32
	o4-mini [76]	2.98	0.61	2.89	2.93	2.36
	Xingchen-Plus [80]	3.12	0.64	3.02	3.05	2.48
	Gemini-2.0-flash-preview-no-thinking [73]	3.27	0.69	3.15	3.18	2.59
	Qwen2.5-7B-Instruct [81]	3.25	0.68	3.10	3.14	2.56
	OmniCharacter-7B [29] (Ours)	3.36	0.72	3.18	3.29	2.65
	UniCharacter-7B (Ours)	3.61	0.80	3.55	3.63	2.91
		3.45	0.63	3.24	3.37	2.69
GPT-4o	GPT-4 [70]	3.12	0.58	3.03	2.92	2.43
	GPT-3.5 [68]	2.90	0.56	2.79	2.86	2.29
	Doubao-1.5-Pro-256K [72]	2.96	0.59	2.86	2.89	2.34
	o4-mini [76]	3.08	0.62	2.99	3.02	2.44
	Xingchen-Plus [80]	3.22	0.67	3.11	3.12	2.54
	Gemini-2.0-flash-preview-no-thinking [73]	3.20	0.66	3.05	3.10	2.52
	Qwen2.5-7B-Instruct [81]	3.33	0.70	3.17	3.26	2.63
	OmniCharacter-7B [29] (Ours)	3.55	0.78	3.49	3.58	2.86
	UniCharacter-7B (Ours)	3.50	0.65	3.30	3.42	2.73
		3.15	0.59	3.06	2.97	2.46
Gemini-1.5-Pro	GPT-4 [70]	2.95	0.58	2.81	2.90	2.32
	GPT-3.5 [68]	2.99	0.61	2.90	2.94	2.37
	Doubao-1.5-Pro-256K [72]	3.13	0.64	3.03	3.06	2.49
	o4-mini [76]	3.26	0.69	3.14	3.15	2.58
	Xingchen-Plus [80]	3.24	0.68	3.08	3.14	2.55
	Gemini-2.0-flash-preview-no-thinking [73]	3.37	0.72	3.20	3.30	2.66
	Qwen2.5-7B-Instruct [81]	3.37	0.72	3.20	3.30	2.66
	OmniCharacter-7B [29] (Ours)	3.58	0.79	3.52	3.60	2.88
	UniCharacter-7B (Ours)	3.50	0.65	3.30	3.42	2.73
		3.15	0.59	3.06	2.97	2.46
Claude-3.5-Sonnet	GPT-4 [70]	2.21	0.28	2.05	2.18	1.67
	GPT-3.5 [68]	2.06	0.21	1.91	1.84	1.48
	Doubao-1.5-Pro-256K [72]	1.74	0.16	1.66	1.70	1.30
	o4-mini [76]	1.79	0.19	1.73	1.76	1.34
	Xingchen-Plus [80]	1.96	0.23	1.87	1.90	1.47
	Gemini-2.0-flash-preview-no-thinking [73]	2.10	0.25	1.99	2.00	1.56
	Qwen2.5-7B-Instruct [81]	2.08	0.24	1.95	1.99	1.54
	OmniCharacter-7B [29] (Ours)	2.17	0.27	2.01	2.09	1.61
	UniCharacter-7B (Ours)	2.36	0.32	2.26	2.34	1.80
		2.23	0.27	2.04	2.16	1.66
Multi-party Dialogue	GPT-4 [70]	2.03	0.20	1.88	1.83	1.45
	GPT-3.5 [68]	1.70	0.15	1.62	1.68	1.27
	Doubao-1.5-Pro-256K [72]	1.76	0.18	1.71	1.73	1.32
	o4-mini [76]	1.93	0.22	1.83	1.86	1.44
	Xingchen-Plus [80]	2.05	0.24	1.96	1.97	1.53
	Gemini-2.0-flash-preview-no-thinking [73]	2.02	0.23	1.91	1.94	1.50
	Qwen2.5-7B-Instruct [81]	2.15	0.26	2.00	2.06	1.59
	OmniCharacter-7B [29] (Ours)	2.32	0.31	2.24	2.31	1.78
	UniCharacter-7B (Ours)	2.24	0.28	2.06	2.17	1.67
		2.05	0.21	1.90	1.85	1.47
GPT-4o	GPT-4 [70]	1.73	0.16	1.65	1.69	1.29
	GPT-3.5 [68]	1.80	0.19	1.72	1.77	1.35
	Doubao-1.5-Pro-256K [72]	1.97	0.23	1.86	1.89	1.46
	o4-mini [76]	2.09	0.25	1.98	1.99	1.55
	Xingchen-Plus [80]	2.07	0.24	1.94	1.97	1.53
	Gemini-2.0-flash-preview-no-thinking [73]	2.19	0.27	2.02	2.10	1.62
	Qwen2.5-7B-Instruct [81]	2.19	0.27	2.02	2.10	1.62
	OmniCharacter-7B [29] (Ours)	2.34	0.32	2.25	2.33	1.79
	UniCharacter-7B (Ours)	2.24	0.28	2.06	2.17	1.67
		2.05	0.21	1.90	1.85	1.47
Gemini-1.5-Pro	GPT-4 [70]	1.73	0.16	1.65	1.69	1.29
	GPT-3.5 [68]	1.80	0.19	1.72	1.77	1.35
	Doubao-1.5-Pro-256K [72]	1.97	0.23	1.86	1.89	1.46
	o4-mini [76]	2.09	0.25	1.98	1.99	1.55
	Xingchen-Plus [80]	2.07	0.24	1.94	1.97	1.53
	Gemini-2.0-flash-preview-no-thinking [73]	2.19	0.27	2.02	2.10	1.62
	Qwen2.5-7B-Instruct [81]	2.19	0.27	2.02	2.10	1.62
	OmniCharacter-7B [29] (Ours)	2.34	0.32	2.25	2.33	1.79
	UniCharacter-7B (Ours)	2.24	0.28	2.06	2.17	1.67
		2.05	0.21	1.90	1.85	1.47
Claude-3.5-Sonnet	GPT-4 [70]	1.73	0.16	1.65	1.69	1.29
	GPT-3.5 [68]	1.80	0.19	1.72	1.77	1.35
	Doubao-1.5-Pro-256K [72]	1.97	0.23	1.86	1.89	1.46
	o4-mini [76]	2.09	0.25	1.98	1.99	1.55
	Xingchen-Plus [80]	2.07	0.24	1.94	1.97	1.53
	Gemini-2.0-flash-preview-no-thinking [73]	2.19	0.27	2.02	2.10	1.62
	Qwen2.5-7B-Instruct [81]	2.19	0.27	2.02	2.10	1.62
	OmniCharacter-7B [29] (Ours)	2.34	0.32	2.25	2.33	1.79
	UniCharacter-7B (Ours)	2.24	0.28	2.06	2.17	1.67
		2.05	0.21	1.90	1.85	1.47

TABLE 7: Prompt perturbation analysis. We evaluate conversation ability under both dyadic and multi-party dialogue settings using different prompt variants. The results show that model performance remains stable across prompt variations.

Model	Dyadic				Multi-party			
	V-1	V-2	V-3	V-4	V-1	V-2	V-3	V-4
GPT-4 [70]	2.76	2.83	2.81	2.80	1.67	1.69	1.66	1.68
GPT-3.5 [68]	2.47	2.46	2.48	2.51	1.48	1.50	1.47	1.49
Doubao-1.5-Pro [72]	2.32	2.30	2.29	2.34	1.30	1.32	1.29	1.31
o4-mini [76]	2.36	2.37	2.30	2.41	1.34	1.35	1.32	1.36
Xingchen-Plus [80]	2.48	2.50	2.55	2.39	1.47	1.49	1.51	1.45
Gemini-2.0-flash-preview-no-thinking [73]	2.59	2.59	2.56	2.53	1.56	1.57	1.54	1.55
Qwen2.5-7B-Instruct [81]	2.56	2.50	2.52	2.54	1.54	1.52	1.53	1.55
OmniCharacter-7B [29] (Ours)	2.65	2.64	2.67	2.65	1.61	1.60	1.62	1.61
UniCharacter-7B (Ours)	2.91	2.88	2.90	2.88	1.80	1.78	1.79	1.81

tween different judges using Pearson correlation coefficients computed on the average model scores. As shown in Tab. 8, the correlations between GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonnet are all above 0.95 with $p < 0.001$, indicating strong agreement across judges and consistent model rankings.

Human Spot-check of GPT-4o Evaluations. We random 100 samples in OmniCharacter++ test set and apply GPT-4o and Human evaluation on the same dimension and prompt.

TABLE 8: Inter-judge agreement analysis. We measure the agreement between different LLM judges using Pearson correlation coefficients (r) computed on the average model scores on overall test set.

Judge Pair	Pearson r	p -value
GPT-4o v.s. Gemini-1.5-Pro	0.99	< 0.001
GPT-4o v.s. Claude-3.5-Sonnet	0.97	< 0.001
Gemini-1.5-Pro v.s. Claude-3.5-Sonnet	0.98	< 0.001

Prompt: Conversation Ability

Given the following information:

- Character Profile: describes the persona, background, and speaking style of the character.
- Dialogue History: the previous turns in the conversation.
- Current Query: the latest user utterance or conversational input.
- Model Response: the response generated by the model.

[V-1: Baseline]

You are an expert evaluator for multi-character role-playing dialogues. Your task is to evaluate the quality of the Model Response based on the following four dimensions:

[V-2: Synonym Substitution]

You are a professional evaluator for multi-character role-playing conversations. Your task is to test the quality of the Model Output based on the following 4 metrics:

[V-3: Word Order Variation]

For multi-character dialogues role-playing, are you an evaluator expert. Your task is to test quality the of the Output Model based on the 4 following metrics:

[V-4: Noise Injection]

You are an () expert evaluator \$\$ for multi-character ~~~~ role-playing dialogues. ## Please evaluate the responses carefully. Your task are to eval the quality of the XXX Response based on the following ~~~~ dimensions:

1. Topic Following (1–5):

Does the response stay relevant to the current query and the ongoing dialogue context? Score 1 if it is off-topic or irrelevant, and 5 if it is fully aligned with the topic.

2. Goal Success (0 or 1):

Does the response successfully achieve the intended dialogue goal or fulfill the instruction? Score 1 if the goal is clearly achieved, otherwise 0.

3. Character Consistency (1–5):

Does the response remain consistent with the assigned character's persona, including tone, style, and behavior? Score 1 if it breaks character, and 5 if it is fully consistent.

4. Dialogue Coherence (1–5):

Does the response appropriately incorporate the dialogue history and maintain logical consistency across turns? Score 1 if it is incoherent, and 5 if it is fully coherent.

Instructions:

- Carefully consider all provided information before making your judgment.
- Evaluate each dimension independently.
- Be strict and avoid overly generous scoring.

Output format:

Provide your evaluation in the following JSON format:

```
{
  "Topic Following":,
  "Goal Success":,
  "Character Consistency":,
  "Dialogue Coherence":
}
```

Fig. 8: Prompt template for conversation ability task.

Fig. 9 shows the distribution of score differences between GPT-4o and human spot-check evaluations (human score-GPT-4o score). As observed, most GPT-4o scores closely match human judgments, with over 72% falling within one standard deviation of the human scores. The distribution is slightly asymmetric, with a larger left region than right, indicating that GPT-4o tends to assign slightly higher scores when disagreements occur.

5.2.3 Human Perception Performance

To comprehensively evaluate speech-language collaboration for immersive role-playing, we conduct human evaluation across six dimensions: fluency, consistency, emotional expression, clarity, appropriateness, and immersion, with scores averaged over five experts (see Tab. 9 and Fig. 10). Baseline models (e.g., SpeechGPT [39], SpeechGPT2 [41], and LLaMA-Omni [37]) show moderate performance, par-

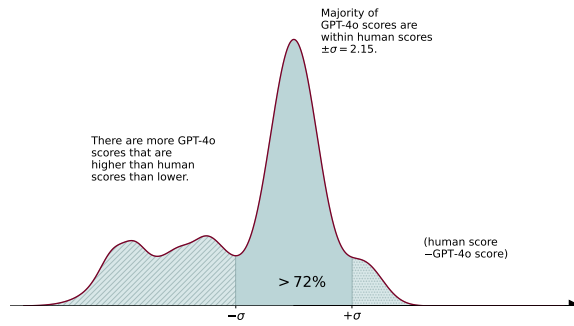


Fig. 9: Distribution of the difference between the scores given by humans and GPT-4o.

TABLE 9: Performance comparison with state-of-the-art models on OmniCharacter++ test set (Human Perception). Human score (averaged from 5 experts) across six voice-related dimensions: fluency (Flu.), consistency (Cons.), emotional expression (Emo.), clarity (Cla.), appropriateness (App.), and immersion (Imm.).

Models	Flu.	Cons.	Emo.	Cla.	App.	Imm.
Dyadic Dialogue						
SpeechGPT [39]	6.58	4.41	4.29	7.51	4.78	4.18
SpeechGPT2 [41]	7.42	5.05	4.88	7.49	5.22	5.33
LLaMA-Omni [37]	7.02	4.53	4.45	6.92	4.62	4.71
OpenOmni [82]	7.26	4.73	4.63	7.23	5.01	4.88
Qwen2.5-Omni [83]	7.41	4.92	3.99	7.22	5.33	5.16
OmniCharacter-7B [29] (Ours)	8.52	7.08	6.61	8.39	6.12	8.97
UniCharacter-7B (Ours)	8.68	7.43	6.94	8.63	6.27	9.18
Multi-party Dialogue						
SpeechGPT [39]	5.48	3.21	3.17	6.20	3.54	2.99
SpeechGPT2 [41]	6.23	3.71	3.64	6.16	4.02	4.01
LLaMA-Omni [37]	5.83	3.36	3.22	5.68	3.41	3.46
OpenOmni [82]	6.01	3.49	3.39	5.99	3.86	3.70
Qwen2.5-Omni [83]	6.19	3.58	2.76	5.94	4.06	3.92
OmniCharacter-7B [29] (Ours)	7.16	5.78	5.31	7.02	4.81	7.58
UniCharacter-7B (Ours)	7.34	6.11	5.59	7.19	4.97	7.79

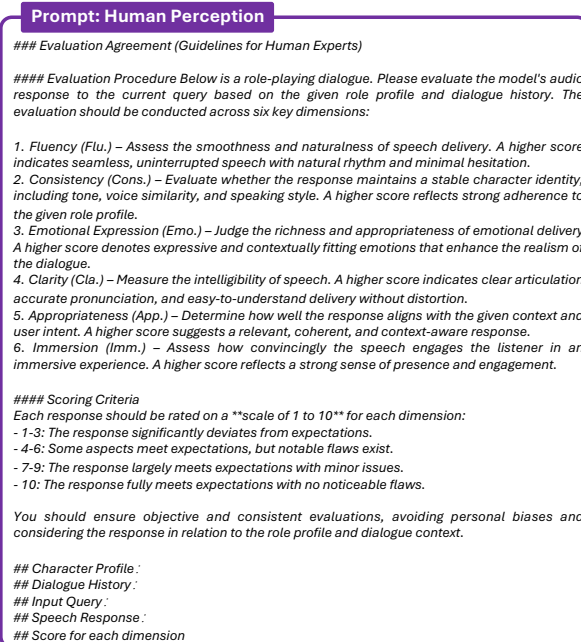


Fig. 10: Prompt template for human perception task.

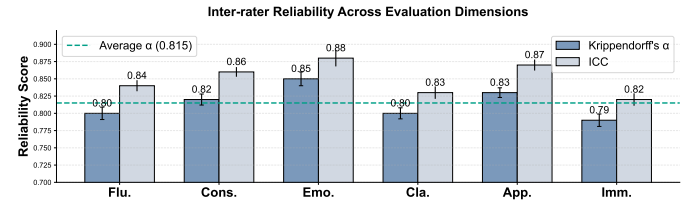


Fig. 11: Inter-rater reliability across evaluation dimensions on OmniCharacter++ test set. Krippendorff's α and Intra-class Correlation Coefficient (ICC) are computed based on ratings from five annotators.

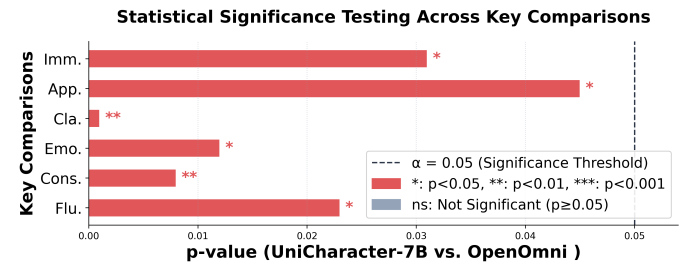


Fig. 12: Statistical significance testing across key evaluation dimensions for model comparison (UniCharacter vs. OpenOmni) on OmniCharacter++ test set.

ticularly struggling with emotional expression and consistency. More recent models, such as OpenOmni [82] and Qwen2.5-Omni [83], exhibit incremental improvements but still face challenges in balancing naturalness and role-specific expressiveness. In contrast, UniCharacter achieves consistently superior performance across all dimensions in both dyadic and multi-party settings, significantly outperforming OmniCharacter and validating the effectiveness of our design. Notably, all models exhibit performance degradation in multi-party dialogue, especially in consistency, emotional expression, and immersion, highlighting the increased difficulty of such scenarios. Overall, our model substantially improves the quality and robustness of realistic, personalized role-playing dialogue in text-to-audio settings.

Inter-rater Reliability. To assess the consistency of human evaluation, we compute Krippendorff's α across five annotators for each evaluation dimension. As shown in Fig. 11, the agreement scores range from 0.79 to 0.85 with an average of 0.815, indicating substantial agreement among annotators. We further report the Intraclass Correlation Coefficient (ICC) as a complementary reliability measure. The consistently high ICC scores across dimensions further validate the robustness of the human evaluation.

Statistical Significance Testing. To validate the effectiveness of model improvements and avoid random sampling bias, we perform statistical significance testing based on human evaluation samples across six core dialogue evaluation dimensions, with all metrics calculated as the average score of all samples per dimension for enhanced reliability. The pairwise comparison results in Fig. 12 demonstrate that the proposed model outperforms the baseline model across most dimensions with statistically significant differences, confirming that the observed performance gaps are not caused by random fluctuations.

TABLE 10: **Ablation study of UniCharacter with different variants on OmniCharacter++ test set under context understanding and conversation ability tasks.** We investigate the effects of using dyadic conversation (Dyadic Conv.) or multi-character conversation (Multi-Conv.) data for training, and also testify the ability of role-contextual dialogue adaptation (RDA) during inference.

Idx	Settings of UniCharacter			Context Understanding										Conversation Ability					
	Dyadic Conv.	Multi-Conv.	RDA	Neg.	Exc.	Free.	Exp.	Inst.	Per.	Conf.	Pla.	Avg.	TF	GS	CC	DC	Avg.		
A	✓	✗	✗	41.28	43.76	68.15	39.52	42.63	41.05	46.84	38.47	45.71	2.41	0.36	2.18	2.33	1.82		
B	✗	✓	✗	41.52	43.91	67.72	38.26	41.58	41.96	47.35	39.12	45.18	2.45	0.39	2.13	2.38	1.84		
C	✓	✓	✗	44.97	47.32	71.08	42.58	45.96	44.51	49.86	41.57	48.23	2.69	0.42	2.51	2.62	2.06		
D	✓	✗	✓	46.21	50.56	73.65	45.12	48.75	47.16	52.49	44.22	51.02	2.91	0.49	2.78	2.88	2.27		
E	✗	✓	✓	46.85	49.14	72.36	43.91	47.23	45.82	51.27	42.94	49.44	2.82	0.46	2.67	2.79	2.19		
F	✓	✓	✓	47.62	50.14	74.82	45.85	48.94	48.12	54.55	44.44	51.81	2.98	0.52	2.89	2.97	2.57		

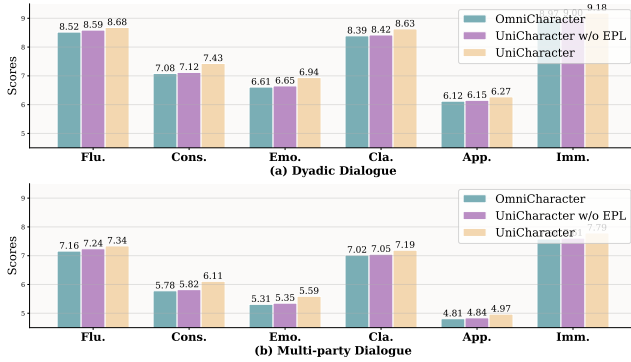


Fig. 13: **Ablation study of EPL module on OmniCharacter++ test set under human perception task.**

5.3 Ablation Studies

To thoroughly investigate individual contributions of each component within our UniCharacter, we conduct a comprehensive ablation study across model variants based on OmniCharacter++ dataset.

Effectiveness of the Designed Components. We conduct a comprehensive ablation study to evaluate the contribution of different components under various settings: ① **Dyadic Conv.**: training only with the dyadic dialogue data; ② **Multi-Conv.**, training only with the multi-character dialogue data, ③ **RDA**: integrating role-contextual dialogue adaptation. The results are summarized in Tab. 10, where each variant is evaluated on both context understanding and conversation ability tasks. We obtain several observations from the table. **First**, each component contributes to performance improvement across two tasks. **Second**, independently training UniCharacter using dyadic/multi-character conversation data (Idx-A/B) brings comparable performance of 45.71% and 45.18% on context understanding and 1.82 and 1.84 on the conversation ability task, while combining both further enhances performance (Idx-C). **Third**, RDA provides additional gains for both settings, validating its effectiveness in test-time adaptation (Idx-D&E). **Finally**, integrating all components achieves the best results (51.81%, 2.57), significantly outperforming other variants. In summary, the results in the table demonstrate the effectiveness of UniCharacter in leveraging diverse training data and test-time adaptation to build higher-quality role-playing interactions.

As discussed in section 4.4, we introduce emotion preference learning (EPL) to enhance emotional speech gen-

eration. We further analyze its impact under both *dyadic* and *multi-party* dialogue settings in Fig. 13. As observed, EPL consistently improves performance across all metrics in both settings. In dyadic dialogue, UniCharacter with EPL achieves clear gains over both OmniCharacter and UniCharacter *w/o* EPL (e.g., Emo.: 6.94 vs. 6.61/6.65). Similar improvements are observed in multi-party dialogue (e.g., Emo.: 5.59 vs. 5.31/5.35), demonstrating that EPL effectively enhances emotional expressiveness and interaction quality.

5.4 Generalization Analysis of UniCharacter

We further evaluate the model's adaptability to distribution shifts by training UniCharacter on the OmniCharacter++ training set and testing on CharacterEval [14], which assesses robustness under out-of-distribution (OOD) characters via a reward model.

5.4.1 Performance on CharacterEval Benchmark

To evaluate generalization to novel characters, we conduct experiments on CharacterEval [14], as shown in Tab. 11, where all characters are out-of-distribution relative to training. We compare our method with both proprietary and open-source role-playing agents, including specialized RPA models [78], [79], as well as larger-scale open-source LLMs [81], [86]. The results show that OmniCharacter achieves an average score of 3.221, outperforming most proprietary and open-source role-playing agents, including MiniMax of 3.134 and InternLM-20B of 3.123, with notable gains in character consistency and role-playing attractiveness. Building on this foundation, UniCharacter further pushes the average score to 3.304, surpassing MiniMax by +0.170 and InternLM-20B by +0.181, despite its smaller scale (7B). These results demonstrate strong cross-character generalization of the proposed methods.

5.4.2 Effectiveness of the Audio Modality

To assess the impact of audio modality, we ablate audio input in OmniCharacter and UniCharacter and evaluate on CharacterEval. Results in Tab. 12 show consistent improvements across all metrics when audio is incorporated. Notably, UniCharacter with audio achieves 3.102 in character consistency and 3.612 in conversational ability, outperforming its audio-ablated variant (3.015 and 3.498). Similar gains are observed for OmniCharacter, indicating strong audio-text synergy in enhancing character fidelity and interaction quality. Furthermore, UniCharacter consistently outperforms OmniCharacter when audio is included, demonstrating its superiority.

TABLE 11: **Quantitative results on the generalizability of state-of-the-art models and our UniCharacter on CharacterEval dataset.** We evaluate across three dimensions including: (A) **Character Consistency**: (a-1) KE: Knowledge-Exposure, (a-2) KA: Knowledge-Accuracy, (a-3) KH: Knowledge-Hallucination, (a-4) PB: Persona-Behavior, (a-5) PU: Persona-Utterance. (B) **Conversational Ability**: (b-1) Flu.: Fluency, (b-2) Coh.: Coherency, (b-3) Cons.: Consistency. (C) **Role-playing Attractiveness**: (c-1) HL: Human-Likeness, (c-2) CS: Communication Skill, (c-3) ED: Expression Diversity, (c-4) Emp.: Empathy. † means we finetune pretrained Qwen-2.5-7B-Instruct [81] on the textual portion of OmniCharacter++ train set.

Models	Character Consistency						Conversational Ability				Role-playing Attractiveness					Avg.
	KE	KA	KH	PB	PU	Avg.	Flu.	Coh.	Cons.	Avg.	HL	CS	ED	Emp.	Avg.	
<i>Proprietary Models</i>																
Baichuan-NPC [78]	1.802	2.964	2.993	2.910	3.151	2.764	3.578	3.898	3.916	3.798	3.836	2.643	2.336	2.971	2.946	3.169
MiniMax [79]	1.835	2.910	2.944	2.774	3.125	2.718	3.609	3.932	3.811	3.784	3.768	2.672	2.150	3.017	2.902	3.134
GPT-3.5 [68]	1.716	2.339	2.212	1.921	2.316	2.101	2.629	2.917	2.700	2.749	2.565	2.422	1.660	2.526	2.293	2.381
GPT-4 [70]	2.250	2.855	2.785	2.721	2.873	2.697	3.332	3.669	3.343	3.448	3.143	3.184	2.153	3.010	2.873	3.006
<i>Open-sourced Models</i>																
ChatGLM3-6B [85]	2.016	2.792	2.704	2.455	2.812	2.556	3.269	3.647	3.283	3.399	3.064	2.932	1.969	2.993	2.739	2.898
Baichuan2-7B [78]	1.813	2.849	2.929	2.830	3.081	2.700	3.551	3.894	3.827	3.757	3.670	2.728	2.115	2.984	2.874	3.110
Baichuan2-13B [78]	1.802	2.869	2.946	2.808	3.081	2.701	3.596	3.924	3.864	3.759	3.700	2.703	2.136	3.021	2.890	3.116
InternLM-7B [86]	1.782	2.800	2.781	2.719	3.016	2.620	3.527	3.823	3.744	3.698	3.546	2.622	2.070	2.897	2.784	2.983
InternLM-20B [86]	1.945	2.916	2.920	2.753	3.041	2.715	3.576	3.943	3.717	3.745	3.582	2.885	2.132	3.047	2.911	3.123
CharacterGLM [17]	1.640	2.819	2.738	2.301	2.969	2.493	3.414	3.717	3.737	3.623	3.738	2.265	1.966	2.812	2.695	2.937
Llama-3.1-8B [74]	2.197	2.701	2.615	3.130	2.704	2.669	3.059	3.477	3.071	3.202	2.922	2.934	2.634	2.759	2.812	2.894
Qwen-7B [87]	1.956	2.728	2.633	2.605	2.780	2.540	3.187	3.564	3.229	3.327	3.036	2.791	2.052	2.838	2.679	2.848
Qwen-14B [87]	1.988	2.800	2.811	2.744	2.900	2.649	3.351	3.765	3.510	3.542	3.354	2.871	2.237	2.970	2.858	3.016
Qwen2-7B-Instruct [75]	1.966	2.537	2.412	2.313	2.436	2.333	2.864	3.171	2.743	2.926	2.655	2.612	1.867	2.654	2.447	2.569
Qwen2.5-7B-Instruct [81]	2.207	2.740	2.633	2.700	2.614	2.579	3.125	3.401	2.971	3.166	2.786	2.871	2.180	2.826	2.666	2.804
OmniCharacter-7B [29] (Ours)	2.230	3.040	2.918	3.531	2.988	2.941	3.369	3.768	3.410	3.516	3.374	3.261	3.002	3.187	3.206	3.221
UniCharacter-7B (Ours)	1.897	3.083	2.998	3.210	3.349	2.907	3.658	4.023	4.122	3.394	3.958	2.821	2.319	3.188	3.072	3.304

TABLE 12: **Ablation study of model training with or without audio modality.** For simplicity, we only report the average score across three dimensions on CharacterEval.

Models	Character Consistency	Conversational Ability	Role-playing Attractiveness
OmniCharacter-7B w/o audio [29]	2.928	3.522	3.168
OmniCharacter-7B [29]	2.941	3.516	3.206
UniCharacter-7B w/o audio	3.015	3.498	3.201
UniCharacter-7B	3.102	3.612	3.295

6 LIMITATION DISCUSSION

Despite its strong performance, OmniCharacter++ has several limitations. First, the dataset relies on neural text-to-speech synthesis, which may introduce artifacts such as limited emotional nuance, pronunciation, and unnatural prosody. Second, although the benchmark evaluates role consistency and incorporates the RDA module, its effectiveness may degrade in large-party dialogues with many unseen and diverse roles. Third, while covering various interactions, long-context scenarios remain challenging due to potential role drift. Finally, the benchmark focuses on open-domain dialogue, leaving domain-specific performance underexplored. Addressing these issues is crucial for building robust and scalable RPAs.

7 CONCLUSION

In this work, we identify a key limitation of current role-playing paradigms, namely the reliance on static interactions and simplified dialogue structures. To address this, we introduce **OmniCharacter++**, the first benchmark for multi-modal, multi-character role-playing, featuring 118K+ multi-turn dialogues, 10K+ diverse characters, and 1M+ synthesized audio responses across 8 topics and 31 subfields. We further propose a comprehensive evaluation framework to assess role-playing agents from multiple perspectives. To enable more realistic interactions, we develop **UniCharacter**, a unified speech-language model with controllable emo-

tional synthesis and in-context learning for multi-character scenarios. Extensive experiments on OmniCharacter++ reveal several innovative findings, which establish a challenging benchmark for state-of-the-art models, paving the way for subsequent advances in role-playing agent research.

REFERENCES

- [1] M. Taylor, A. B. Sachet, B. L. Maring, and A. M. Mannering, "The assessment of elaborated role-play in young children: Invisible friends, personified objects, and pretend identities," *Social Development*, vol. 22, no. 1, pp. 75–93, 2013.
- [2] A. Bowman, "Companionship games: a framework for emotionally engaging and empathetic interactive characters," in *ICIDS*. Springer, 2019, pp. 413–417.
- [3] T. Duus Henriksen *et al.*, "Moving educational role-play beyond entertainment," 2010.
- [4] D. Craciun, "Role-playing as a creative method in science education," *Journal of Science and Arts*, vol. 10, no. 1, p. 175, 2010.
- [5] N. Afonso and R. Prada, "Agents that relate: Improving the social believability of non-player characters in role-playing games," in *ICEC*. Springer, 2008, pp. 34–45.
- [6] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *UIST*, 2023, pp. 1–22.
- [7] J. Li, Z. Jie, X. Wang, X. Wei, and L. Ma, "Expansion and shrinkage of localization for weakly-supervised semantic segmentation," *NeurIPS*, vol. 35, pp. 16 037–16 051, 2022.
- [8] L. Wang, J. Lian, Y. Huang, Y. Dai, H. Li, X. Chen, X. Xie, and J.-R. Wen, "Characterbox: Evaluating the role-playing capabilities of llms in text-based virtual worlds," in *NAACL*, 2025, pp. 6372–6391.
- [9] N. Wang, Z. Peng, H. Que, J. Liu, W. Zhou, Y. Wu, H. Guo, R. Gan, Z. Ni, J. Yang *et al.*, "Rolellm: Benchmarking, eliciting, and enhancing role-playing abilities of large language models," in *ACL (Findings)*, 2024, pp. 14 743–14 777.
- [10] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-llm: A trainable agent for role-playing," in *EMNLP*, 2023, pp. 13 153–13 187.
- [11] M. Kosinski, "Evaluating large language models in theory of mind tasks," *PNAS*, vol. 121, no. 45, p. e2405460121, 2024.
- [12] T. R. Goldstein and E. Winner, "Engagement in role play, pretense, and acting classes predict advanced theory of mind skill in middle childhood," *Imagination, Cognition and Personality*, vol. 30, no. 3, pp. 249–258, 2011.
- [13] T. Ge, X. Chan, X. Wang, D. Yu, H. Mi, and D. Yu, "Scaling synthetic data creation with 1,000,000,000 personas," *arXiv preprint arXiv:2406.20094*, 2024.

- [14] Q. Tu, S. Fan, Z. Tian, T. Shen, S. Shang, X. Gao, and R. Yan, "CharacterEval: A Chinese benchmark for role-playing conversational agent evaluation," in *ACL*, 2024, pp. 11 836–11 850.
- [15] C. Li, Z. Leng, C. Yan, J. Shen, H. Wang, W. Mi, Y. Fei, X. Feng, S. Yan, H. Wang *et al.*, "Chatharuhi: Reviving anime character in reality via large language model," *arXiv preprint arXiv:2308.09597*, 2023.
- [16] N. Chen, Y. Wang, H. Jiang, D. Cai, Y. Li, Z. Chen, L. Wang, and J. Li, "Large language models meet Harry Potter: A dataset for aligning dialogue agents with characters," in *EMNLP*, 2023, pp. 8506–8520.
- [17] J. Zhou, Z. Chen, D. Wan, B. Wen, Y. Song, J. Yu, Y. Huang, P. Ke, G. Bi, L. Peng *et al.*, "CharacterGLM: Customizing social characters with large language models," in *EMNLP*, 2024, pp. 1457–1476.
- [18] T. Gosling, A. Dale, and Y. Zheng, "Pippa: A partially synthetic conversational dataset," *arXiv preprint arXiv:2308.05884*, 2023.
- [19] T. Shen, S. Li, and D. Xiong, "RoleEval: A bilingual role evaluation benchmark for large language models," 2023.
- [20] K. Lu, B. Yu, C. Zhou, and J. Zhou, "Large language models are superpositions of all characters: Attaining arbitrary role-play via self-alignment," in *ACL*, 2024, pp. 7828–7840.
- [21] Y. Ran, X. Wang, R. Xu, X. Yuan, J. Liang, Y. Xiao, and D. Yang, "Capturing minds, not just words: Enhancing role-playing language models with personality-indicative data," in *EMNLP (Findings)*, 2024, pp. 14 566–14 576.
- [22] Y. Tang, J. Ou, C. Liu, F. Zhang, D. Zhang, and K. Gai, "Enhancing role-playing systems through aggressive queries: Evaluation and improvement," *arXiv preprint arXiv:2402.10618*, 2024.
- [23] X. Yuan, S. Yuan, Y. Cui, T. Lin, X. Wang, R. Xu, J. Chen, and D. Yang, "Evaluating character understanding of large language models via character profiling from fictional works," in *EMNLP*, 2024, pp. 8015–8036.
- [24] H. Chen, H. Chen, M. Yan, W. Xu, G. Xing, W. Shen, X. Quan, C. Li, J. Zhang, and F. Huang, "SocialBench: Sociality evaluation of role-playing conversational agents," in *ACL (Findings)*, 2024, pp. 2108–2126.
- [25] J. Ahn, T. Lee, J. Lim, J.-H. Kim, S. Yun, H. Lee, and G. Kim, "TimeChara: Evaluating point-in-time character hallucination of role-playing large language models," in *ACL (Findings)*, 2024, pp. 3291–3325.
- [26] R. Xu, X. Wang, J. Chen, S. Yuan, X. Yuan, J. Liang, Z. Chen, X. Dong, and Y. Xiao, "Character is destiny: Can large language models simulate persona-driven decisions in role-playing?" *arXiv e-prints*, pp. arXiv-2404, 2024.
- [27] X. Wang, Y. Xiao, J.-t. Huang, S. Yuan, R. Xu, H. Guo, Q. Tu, Y. Fei, Z. Leng, W. Wang *et al.*, "InCharacter: Evaluating personality fidelity in role-playing agents through psychological interviews," in *ACL*, 2024, pp. 1840–1873.
- [28] Y. Dai, H. Hu, L. Wang, S. Jin, X. Chen, and Z. Lu, "Mmrole: A comprehensive framework for developing and evaluating multi-modal role-playing agents," in *ICLR*, 2025.
- [29] H. Zhang, R. Luo, X. Liu, Y. Wu, T.-E. Lin, P. Zeng, Q. Qu, F. Fang, M. Yang, L. Gao, J. Song, F. Huang, and Y. Li, "Omnicharacter: Towards immersive role-playing agents with seamless speech-language personality interaction," in *ACL*, 2025, pp. 26 318–26 331.
- [30] N. Sadeq, Z. Xie, B. Kang, P. Lamba, X. Gao, and J. J. McAuley, "Mitigating hallucination in fictional character role-play," in *EMNLP (Findings)*, 2024, pp. 14 467–14 479.
- [31] Z. M. Wang, Z. Peng, H. Que, J. Liu, W. Zhou, Y. Wu, H. Guo, R. Gan, Z. Ni, J. Yang *et al.*, "RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models," *arXiv preprint arXiv:2310.00746*, 2023.
- [32] F. Fang, T.-E. Lin, Y. Wu, X. Liu, X. Huang, D. Chen, J. Ye, H. Zhang, L. Zhu, H. Alinejad-Rokny *et al.*, "Charm: Character-based act-adaptive reward modeling for advanced role-playing language agents," *arXiv preprint arXiv:2505.23923*, 2025.
- [33] S. Wang, M. Zhang, D. Zhang, A. Belatreche, Y. Xiao, Y. Liang, Y. Shan, Q. Sun, E. Zhang, and Y. Yang, "Spiking vision transformer with saccadic attention," in *ICLR*, 2025.
- [34] S. Wang, D. Zhang, A. Belatreche, Y. Xiao, H. Qing, W. Wei, M. Zhang, and Y. Yang, "Ternary spike-based neuromorphic signal processing system," *Neural Networks*, vol. 187, p. 107333, 2025.
- [35] Y. Xiao, S. Wang, D. Zhang, W. Wei, Y. Shan, X. Liu, Y. Jiang, and M. Zhang, "Rethinking spiking self-attention mechanism: Implementing a-xnor similarity calculation in spiking transformers," in *CVPR*, 2025, pp. 5444–5454.
- [36] Y. Liu, H. Duan, Y. Zhang, B. Li, S. Zhang, W. Zhao, Y. Yuan, J. Wang, C. He, Z. Liu *et al.*, "Mmbench: Is your multi-modal model an all-around player?" in *ECCV*, 2024, pp. 216–233.
- [37] Q. Fang, S. Guo, Y. Zhou, Z. Ma, S. Zhang, and Y. Feng, "Llama-omni: Seamless speech interaction with large language models," *arXiv preprint arXiv:2409.06666*, 2024.
- [38] T. Wang, J. Yi, R. Fu, C. Qiang, D. Chong, C. Wang, D. Dai, Z. Wen, and J. Tao, "Speechpalette: A comprehensive speech editing method for text-based speech editing, one-shot tts and attributes editing," *TPAMI*, 2025.
- [39] D. Zhang, S. Li, X. Zhang, J. Zhan, P. Wang, Y. Zhou, and X. Qiu, "Speechgpt: Empowering large language models with intrinsic cross-modal conversational abilities," in *EMNLP (Findings)*, 2023, pp. 15 757–15 773.
- [40] A. Zeng, Z. Du, M. Liu, K. Wang, S. Jiang, L. Zhao, Y. Dong, and J. Tang, "GLM-4-voice: Towards intelligent and human-like end-to-end spoken chatbot," *arXiv preprint arXiv:2412.02612*, 2024.
- [41] D. Zhang, X. Zhang, J. Zhan, S. Li, Y. Zhou, and X. Qiu, "Speechgpt-gen: Scaling chain-of-information speech generation," *arXiv preprint arXiv:2401.13527*, 2024.
- [42] T. Wang, L. Zhou, Z. Zhang, Y. Wu, S. Liu, Y. Gaur, Z. Chen, J. Li, and F. Wei, "Viola: Conditional language models for speech recognition, synthesis, and translation," *TASLP*, 2024.
- [43] Z. Borsos, R. Marinier, D. Vincent, E. Kharitonov, O. Pietquin, M. Sharifi, D. Roblek, O. Teboul, D. Grangier, M. Tagliasacchi, and N. Zeghidour, "Audiolm: A language modeling approach to audio generation," *TASLP*, pp. 2523–2533, 2023.
- [44] C. Wang, S. Chen, Y. Wu, Z. Zhang, L. Zhou, S. Liu, Z. Chen, Y. Liu, H. Wang, J. Li *et al.*, "Neural codec language models are zero-shot text to speech synthesizers," *arXiv preprint arXiv:2301.02111*, 2023.
- [45] P. K. Rubenstein, C. Asawaroengchai, D. D. Nguyen, A. Bapna, Z. Borsos, F. d. C. Quirry, P. Chen, D. E. Badawy, W. Han, E. Kharitonov *et al.*, "Audiopalm: A large language model that can speak and listen," *arXiv preprint arXiv:2306.12925*, 2023.
- [46] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour, "Moshi: a speech-text foundation model for real-time dialogue," *arXiv preprint arXiv:2410.00037*, 2024.
- [47] Q. Zhang, L. Cheng, C. Deng, Q. Chen, W. Wang, S. Zheng, J. Liu, H. Yu, C.-H. Tan, Z. Du, and S. Zhang, "Omniflatten: An end-to-end gpt model for seamless voice conversation," in *ACL*, 2025, pp. 14 570–14 580.
- [48] S. Ding, W. Du, L. Ding, J. Zhang, L. Guo, and B. An, "Robust multi-agent communication with graph information bottleneck optimization," *TPAMI*, vol. 46, no. 5, pp. 3096–3107, 2023.
- [49] C.-M. Chan, W. Chen, Y. Su, J. Yu, W. Xue, S. Zhang, J. Fu, and Z. Liu, "Chateval: Towards better llm-based evaluators through multi-agent debate," *arXiv preprint arXiv:2308.07201*, 2023.
- [50] Q. Sun, Z. Yin, X. Li, Z. Wu, X. Qiu, and L. Kong, "Corex: Pushing the boundaries of complex reasoning through multi-model collaboration," *arXiv preprint arXiv:2310.00280*, 2023.
- [51] T. Hu, B. Luo, C. Yang, and T. Huang, "Mo-mix: Multi-objective multi-agent cooperative decision-making with deep reinforcement learning," *TPAMI*, vol. 45, no. 10, pp. 12 098–12 112, 2023.
- [52] R. Liu, R. Yang, C. Jia, G. Zhang, D. Zhou, A. M. Dai, D. Yang, and S. Vosoughi, "Training socially aligned language models in simulated human society," in *ICLR*, 2024.
- [53] J. Chen, X. Wang, R. Xu, S. Yuan, Y. Zhang, W. Shi, J. Xie, S. Li, R. Yang, T. Zhu *et al.*, "From persona to personalization: A survey on role-playing language agents," *arXiv preprint arXiv:2404.18231*, 2024.
- [54] P. Gunkel. (1998) Human kaleidoscope. [Online]. Available: <https://ideonomy.mit.edu/essays/kaleidoscope.html>
- [55] Sixseconds. (2022) Plutchik's wheel of emotions. [Online]. Available: <https://www.6seconds.org/2025/02/06/plutchik-wheel-emotions/>
- [56] Z. Du, Q. Chen, S. Zhang, K. Hu, H. Lu, Y. Yang, H. Hu, S. Zheng, Y. Gu, Z. Ma *et al.*, "Cosyvoice: A scalable multilingual zero-shot text-to-speech synthesizer based on supervised semantic tokens," *arXiv preprint arXiv:2407.05407*, 2024.
- [57] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *ICML*, 2023, pp. 28 492–28 518.
- [58] S. Hu, L. Zhou, S. Liu, S. Chen, L. Meng, H. Hao, J. Pan, X. Liu, J. Li, S. Sivasankaran, L. Liu, and F. Wei, "Wavlm: Towards robust and adaptive speech large language model," in *EMNLP (Findings)*, 2024, pp. 4552–4572.

- [59] J. Li, M. Galley, C. Brockett, J. Gao, and W. B. Dolan, "A diversity-promoting objective function for neural conversation models," in *NAACL*, 2016, pp. 110–119.
- [60] Y. Zhu, S. Lu, L. Zheng, J. Guo, W. Zhang, J. Wang, and Y. Yu, "Texygen: A benchmarking platform for text generation models," in *SIGIR*, 2018, pp. 1097–1100.
- [61] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei *et al.*, "Qwen2. 5 technical report," *arXiv preprint arXiv:2412.15115*, 2024.
- [62] Z. Du, Y. Wang, Q. Chen, X. Shi, X. Lv, T. Zhao, Z. Gao, Y. Yang, C. Gao, H. Wang *et al.*, "Cosyvoice 2: Scalable streaming speech synthesis with large language models," *arXiv preprint arXiv:2412.10117*, 2024.
- [63] Y. Chu, J. Xu, Q. Yang, H. Wei, X. Wei, Z. Guo, Y. Leng, Y. Lv, J. He, J. Lin *et al.*, "Qwen2-audio technical report," *arXiv preprint arXiv:2407.10759*, 2024.
- [64] K. Chen, Y. Gou, R. Huang, Z. Liu, D. Tan, J. Xu, C. Wang, Y. Zhu, Y. Zeng, K. Yang *et al.*, "Emova: Empowering language models to see, hear and speak with vivid emotions," in *CVPR*, 2025, pp. 5455–5466.
- [65] J. Zhan, J. Dai, J. Ye, Y. Zhou, D. Zhang, Z. Liu, X. Zhang, R. Yuan, G. Zhang, L. Li *et al.*, "Anygpt: Unified multimodal llm with discrete sequence modeling," in *ACL*, 2024, pp. 9637–9662.
- [66] M. L. Menéndez, J. A. Pardo, L. Pardo, and M. d. C. Pardo, "The jensen-shannon divergence," *Journal of the Franklin Institute*, vol. 334, no. 2, pp. 307–318, 1997.
- [67] J. Kong, J. Kim, and J. Bae, "Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis," *NeurIPS*, vol. 33, pp. 17 022–17 033, 2020.
- [68] OpenAI, "Gpt-3.5 turbo fine-tuning and api updates," 2023. [Online]. Available: <https://openai.com/index/gpt-3-5-turbo-fine-tuning-and-api-updates/>
- [69] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.
- [70] OpenAI, "Introducing gpt-4.1 in the api," 2025. [Online]. Available: <https://openai.com/index/gpt-4-1>
- [71] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan *et al.*, "Deepseek-v3 technical report," *arXiv preprint arXiv:2412.19437*, 2024.
- [72] D. Guo, F. Wu, F. Zhu, F. Leng, G. Shi, H. Chen, H. Fan, J. Wang, J. Jiang, J. Wang *et al.*, "Seed1. 5-v1 technical report," *arXiv preprint arXiv:2505.07062*, 2025.
- [73] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican *et al.*, "Gemini: a family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2023.
- [74] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan *et al.*, "The llama 3 herd of models," *arXiv e-prints*, pp. arXiv-2407, 2024.
- [75] Q. Team, "Qwen2 technical report," *arXiv preprint arXiv:2407.10671*, 2024.
- [76] OpenAI, "Openai o3 and o4-mini system card," 2025. [Online]. Available: <https://openai.com/research/o3-o4-mini-system-card>
- [77] A. Jaech, A. Kalai, A. Lerer, A. Richardson, A. El-Kishky, A. Low, A. Helyar, A. Madry, A. Beutel, A. Carney *et al.*, "Openai o1 system card," *arXiv preprint arXiv:2412.16720*, 2024.
- [78] A. Yang, B. Xiao, B. Wang, B. Zhang, C. Bian, C. Yin, C. Lv, D. Pan, D. Wang, D. Yan *et al.*, "Baichuan 2: Open large-scale language models," *arXiv preprint arXiv:2309.10305*, 2023.
- [79] A. Li, B. Gong, B. Yang, B. Shan, C. Liu, C. Zhu, C. Zhang, C. Guo, D. Chen, D. Li *et al.*, "Minimax-01: Scaling foundation models with lightning attention," *arXiv preprint arXiv:2501.08313*, 2025.
- [80] T. Lab, "Tongyi xingcheng plus," 2023. [Online]. Available: <https://tongyi.aliyun.com/xingchen>
- [81] Q. Team, "Qwen2.5: A party of foundation models," September 2024.
- [82] R. Luo, T.-E. Lin, H. Zhang, Y. Wu, X. Liu, M. Yang, Y. Li, L. Chen, J. Li, L. Zhang *et al.*, "Openomni: Large language models pivot zero-shot omnimodal alignment across language with real-time self-aware emotional speech synthesis," *NeurIPS*, 2025.
- [83] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang *et al.*, "Qwen2. 5-omni technical report," *arXiv preprint arXiv:2503.20215*, 2025.
- [84] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [85] T. GLM, A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Zhang, D. Rojas, G. Feng, H. Zhao *et al.*, "Chatglm: A family of large language models from glm-130b to glm-4 all tools," *arXiv preprint arXiv:2406.12793*, 2024.
- [86] I. Team, "Internlm: A multilingual language model with progressively enhanced capabilities," 2023.
- [87] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang *et al.*, "Qwen technical report," *arXiv preprint arXiv:2309.16609*, 2023.

8 BIOGRAPHY SECTION

Haonan Zhang received the B.E. degree in computer science and technology from Xidian University in 2020. He is currently pursuing the Ph.D. degree in computer science and technology from the University of Electronic Science and Technology of China. His research interests include Multimodal Understanding and Machine Learning.

Pengpeng Zeng received the B.E. degree from Xi'an University of Technology in 2016, and the M.E. and Ph.D. degrees from University of Electronic Science and Technology of China in 2019 and 2023, respectively. He is now a researcher in Tongji University, China. His current research interests include visual understanding, machine learning, and reinforcement learning.

Ji Zhang is an Assistant Professor with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, China. He obtained his PhD degree from University of Electronic Science and Technology of China in 2024, under the supervision of Prof. Jingkuan Song. His research interests include few-shot learning, transfer learning and robotics. He has published over ten papers on top conferences/journals, such as CVPR'24-25, ICCV'23, IJML'23, TIP'23.

Jingkuan Song is a professor with the School of Computer Science and Technology, Tongji University, China. He joined Columbia University as a Postdoc Research Scientist (2016-2017), and University of Trento as a Research Fellow (2014-2016). He obtained his PhD degree in 2014 from The University of Queensland (UQ), Australia. His research interest includes large-scale multimedia retrieval, LLMs and deep learning techniques. He was the winner of the Best Paper Award in ICPR (2016, Mexico), Best Student Paper Award in Australian Database Conference (2017, Australia), and Best Paper Honorable Mention Award (2017, Japan). He is an Associate Editor of IEEE TMM and ACM TOMM.

Nicu Sebe is a professor with the Department of Information Engineering and Computer Science, University of Trento, leading the research in the areas of multimedia information retrieval and human behavior understanding. He was the General CoChair of ACM Multimedia 2013 and 2022, and the Program Chair of ACM Multimedia 2007 and 2011, ECCV 2016, ICCV 2017 and ICPR 2020. He is a fellow of the International Association for Pattern Recognition (IAPR) and of the European Laboratory for Learning and Intelligent Systems (ELLIS).

Heng Tao Shen is a professor with the School of Computer Science and Technology, Tongji University, China. He obtained his BSc with 1st class Honours and PhD from Department of Computer Science, National University of Singapore in 2000 and 2004 respectively. His current research interests include multimedia search, computer vision, artificial intelligence, and big data management. He has published 300+ peer-reviewed papers and received 7 best paper awards from international conferences, including the Best Paper Award from ACM Multimedia 2017 and Best Paper Award-Honorable Mention from ACM SIGIR 2017. He has served as General Co-chair for ACM Multimedia 2021 and TPC Co-Chair for ACM Multimedia 2015, and is an Associate Editor of ACM Trans. of Data Science (TDS), IEEE Trans. on Image Processing (TIP), IEEE Trans. on Multimedia (TMM), and IEEE Trans. on Knowledge and Data Engineering (TKDE). He is a Fellow of ACM/IEEE/OSA.

Lianli Gao is a professor with the School of Computer Science and Engineering, University of Electronic Science and Technology of China. She obtained her PhD degree in Information Technology from The University of Queensland (UQ), Australia, under the supervision of Prof. Jane Hunter and Prof. Michael Bruenig. Her research ranges from Semantic Web, Machine Learning, Deep Learning, Computer Vision (Images and Videos), NLP, Knowledge Reasoning, Knowledge and the related practical applications etc. Specifically, she is mainly focusing on integrating Natural Language for Visual Content Understanding. She has the winner of the IEEE Transactions on Multimedia 2020 Prize Paper Award, Best Student Paper Award in Australian Database Conference (2017, Australia), IEEE TCMC Rising Star Award 2020 and ALIBABA Academic Young Fellow. She is an Associate Editor of IEEE TMM.